

Design-Weighted Latent Class Analysis of Survey Data: An Adaptable Analysis Script

Institutional trust in Mexico: LAPOP AmericasBarometer 2023

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1 Purpose and how to use this script

This is a complete, adaptable analysis for finding latent classes (hidden respondent types) in categorical survey items collected under a complex sampling design, estimating the class structure so that it describes the **population** rather than the achieved sample, attaching design-based confidence intervals to every estimated probability, writing class membership back to the respondents, and profiling that membership on demographics.

The analysis identifies types of respondents by their **trust in Mexican institutions**, thirteen 7-point items from the 2023 LAPOP AmericasBarometer (1 = Not at all, 7 = A lot). Everything dataset-specific lives in ONE adapter script (`prepare_data_lapop.R`) plus the configuration block; to apply this to another survey, write a sibling adapter and edit the config, and nothing else changes.

The script makes four methodological commitments, each explained where it occurs:

1. **The design is respected during estimation**, not only afterward. Class prevalences and item-response probabilities are estimated by *pseudo-maximum likelihood* with the base weights, which is the analogue of declaring `svyset psu [pweight = w], strata(stratum)` in Stata. This is the established design-based approach to latent class analysis under complex sampling (Patterson, Dayton, and Graubard 2002, *Journal of the American Statistical Association* 97(459):721–728; Vermunt and Magidson 2007, *Sociological Methods & Research* 36(1):87–111).
2. **poLCA is used as an unweighted benchmark**, because it cannot use design weights, and the gap between it and the weighted fit measures how informative the design is for this typology. A built-in equivalence check confirms the weighted estimator reproduces poLCA exactly when all weights are set to one.
3. **Uncertainty is design-based**. Confidence intervals on every conditional and inclusion probability come from the stratified jackknife (JKn) using replicate weights, which respects the stratification and clustering of a multistage design.

4. **“Don’t know” is treated as missing**, not as a response category. The model is fit on respondents who answered every item, which also makes the `poLCA` benchmark an exact comparison; respondents dropped for a missing answer are then assigned a class from their answered items (the scoring section), so no one is discarded and nothing is imputed. The configuration block notes where to change this if your items call for it.

The code uses the tidyverse, contains no `for` or `while` loops (iteration is done with `purrr` and matrix algebra), and is written to fail loudly and informatively rather than silently on edge cases.

2 Setup

```
library(tidyverse)      # data manipulation and plotting
library(matrixStats)    # numerically stable row-wise log-sum-exp and cumulative
  ↪ sums
library(survey)         # complex-design objects, replicate weights, svyglm
library(poLCA)          # unweighted benchmark latent class model
library(clue)           # solve_LSAP: optimal label alignment across fits
library(furrr)          # future_map_*: parallel model fits (enumeration, screen,
  ↪ poLCA)

# All reusable machinery lives in one annotated source file: the plotting and
# data-prep helpers, the weighted EM engine and label alignment, the LLM
# class-labeling functions. Keep
# survey_lca_source.R in the same folder as
# this document; the render fails if it is not on the path. Sourcing it here,
  ↪ after
# the libraries, means every function is available to the analysis chunks below.
source("survey_lca_source.R")

`%||%` <- rlang::`%||%` # null-coalescing helper (also defined in the source
  ↪ file)
# Reproducibility is controlled by cfg$seed (set in the configuration block):
  ↪ every
# random step below (the EM random starts and the poLCA starts) seeds
# itself from it, so the classes and their labels are identical across runs.

# poLCA attaches MASS, whose select() and some matrixStats verbs (count) mask
# their tidyverse equivalents. We namespace-qualify dplyr::select(),
  ↪ dplyr::count(),
# and dplyr::distinct() throughout. This is a deliberate, defensive choice.
```

3 The dataset adapter

One script per dataset does all reading and cleaning; it is sourced here, before the configuration, so the config can point at its outputs (`lapop`, `lapop_items`). Its console output, the nonresponse audit and the preparation summary, prints below and is part of the record.

```
source("prepare_data_lapop.R")
```

Value-label inventory (distinct labels across the 13 items):

```
# A tibble: 5 x 3
  value label n_items
  <dbl> <chr>   <int>
1      1 Nada      13
2      7 Mucho     13
3 888888 DK        13
4 988888 NR        13
5 999999 N/A       13
```

Nonresponse labels recoded to NA:

```
# A tibble: 3 x 2
  value label
  <dbl> <chr>
1 888888 DK
2 988888 NR
3 999999 N/A
```

Recode audit (derived variables; 'unmatched' = source present, recode NA):

```
# A tibble: 7 x 4
  derived      source na_share unmatched
  <chr>      <chr>    <dbl>    <dbl>
1 age_cat    q2         0         0
2 male       q1tc_r     0.003     0
3 Edu        edre      0.001     0
4 Urban      ur         0         0
5 Employment ocup4a     0.003     0
6 satis_demo pn4       0.018     0
7 prez_rating m1        0.003     0
```

Attrition: 1622 read -> 1580 prepared (42 dropped by the listwise rule on design/demographics/
Prepared: 1580 respondents, 13 items (1..7), 150 item-partial cases kept for scoring.

4 Configuration: the only block you edit to reuse this script

Everything design-specific is gathered here; the data themselves arrive already prepared from the adapter script sourced in the previous section. *Why this block matters: it is the single point of*

contact between you and the analysis. Every section downstream reads from `cfg`, so a correct configuration here is what makes the rest of the script reusable without edits.

```
cfg <- list(
  # --- columns in YOUR data -----
  items = lapop_items,      # the 13 trust items (English names from the
    ↪ adapter)
  aux    = c("age_cat", "male", "Edu", "Urban", "Employment",
    "satis_demo", "prez_rating"),      # demographics/outcomes for
    ↪ profiling
  strata = "strata",        # stratum identifier (design)
  psu     = "upm",          # primary sampling unit / cluster identifier
    ↪ (design)
  weight  = "wt",           # base weight (design); constant 1 in this file,
    ↪ so the
                                # weighted machinery reduces to unweighted,
                                ↪ correctly

  # --- analysis choices -----
  K_range = 2:16,           # candidate numbers of latent classes to compare
  K_force = 5,              # force a specific number of classes, overriding the
    ↪ BIC pick;
                                # leave NULL to select K by BIC (Section 8). Set
                                ↪ to an integer
                                # within K_range to pin K (e.g. when BIC prefers
                                ↪ more classes
                                # than are interpretable). The enumeration still
                                ↪ prints either way.
  n_starts = 20L,          # random starts per fit (guards against local
    ↪ maxima)
  seed = 2026L,             # master random seed: fix it and the classes (and
    ↪ their
                                # labels) are identical on every run of the same
                                ↪ data
  parallel = TRUE,          # fit the candidate-K models, the indicator
    ↪ screen, and the
                                # poLCA sweep in parallel via furrr. Output is
                                ↪ identical to
                                # serial because every fit re-seeds itself from
                                ↪ `seed`.
  workers = NULL,          # parallel workers; NULL uses availableCores() -
    ↪ 1. Ignored
                                # when parallel = FALSE.
  na_codes = NULL,         # response code(s) to treat as MISSING, e.g. 9
    ↪ for "Don't know";
                                # leave NULL if your data already uses NA for
                                ↪ missing/DK
```

```

# --- LLM class labeling -----
class_labels_csv = NULL,      # analyst label file (columns K, Label,
  ↳ Description).

                                # Requires K_force; when set, the LLM never runs.
compass_base_url = "https://openrouter.ai/api/v1", # OpenAI-compatible
  ↳ endpoint: OpenRouter

                                # at home; at work replace with your compass
                                ↳ endpoint,
                                # (or OpenRouter at home). The API key is
                                ↳ supplied by
                                # your .Rprofile; nothing is stored in this
                                ↳ script.
llm_model          = "google/gemma-4-31b-it", # model at that endpoint (work:
  ↳ your model name)
llm_key_env        = "OPENROUTER_API_KEY",    # env var holding the API key
  ↳ (.Renviro). At

                                # work set NULL: ellmer then uses OPENAI_API_KEY
                                ↳ from .Rprofile.
survey_context     = paste("The 2023 AmericasBarometer (LAPOP) survey of adults
  ↳ in",

                                "Mexico; items measure trust in national
  ↳ institutions."),
                                # one-sentence context (country, year, topic) so
                                ↳ the
                                # labeler can resolve what the items refer to.
                                ↳ Rule 1 fences it
                                # to referent resolution only. NULL keeps the
                                ↳ certified
                                # context-free prompt byte for byte.

# --- outputs -----
out_dir = "outputs",          # folder where every result file is written and
  ↳ where the

                                # frozen label cache (class_labels_llm.csv) is
                                ↳ found on
                                # re-render; created below if missing

# --- data -----
data = lapop                  # the prepared tibble from the adapter script;
  ↳ swap the

                                # sourced adapter (and this line) to change
                                ↳ datasets
)
dir.create(cfg$out_dir, showWarnings = FALSE, recursive = TRUE)

# Fail fast on a missing API key: quarto renders in a FRESH session (console
# Sys.setenv() does not reach it), so if labeling will need the LLM this run

```

```
# (no analyst CSV, no frozen cache), verify the key NOW, in seconds, instead
# of after the enumeration has burned its minutes.
needs_llm <- is.null(cfg$class_labels_csv) &&
  !file.exists(file.path(cfg$out_dir, "class_labels_llm.csv"))
if (needs_llm && !nzchar(Sys.getenv(cfg$llm_key_env %||% "OPENAI_API_KEY")))
  stop("Class labeling needs the LLM this run, but the API key env var '",
       cfg$llm_key_env %||% "OPENAI_API_KEY", "' is empty. Put the key in ",
       ".Renviron (home) or let .Rprofile set it (work); console Sys.setenv() ",
       "does not reach a quarto render.")
```

A note on the **items coding requirement**, which the script checks: each indicator must be a small set of consecutive integer codes starting at 1 over its *substantive* responses (for example 1,2,3,4 for a four-point item), and items may differ in how many categories they have. “Don’t know” is **not** a response category in this version: it is treated as missing data, set to NA (directly, or by listing its code in `cfg$na_codes`). Respondents with any missing item are held out of the model fit (Section 6) but are still assigned a class afterwards from their answered items (the scoring section), so no one is discarded and nothing is imputed.

```
# theme_lca() (the colorblind-safe figure theme) lives in survey_lca_source.R.
# Only the two-color weighted/unweighted palette, which is specific to this
# document's labels, is defined here.
wu_pal <- setNames(viridisLite::viridis(2, begin = 0.2, end = 0.75),
                  c("Weighted (population)", "Unweighted (poLCA)"))
```

```
# Parallel backend for the repeated model fits below (enumeration, indicator
# screen, poLCA sweep), read from cfg. multisession works on Windows and macOS.
# A sequential plan reproduces the serial result exactly, because every fit
# re-seeds itself from cfg$seed; parallelism changes timing, not output.
if (isTRUE(cfg$parallel)) {
  future::plan(future::multisession,
               workers = cfg$workers %||% max(1L, future::availableCores() - 1L))
} else {
  future::plan(future::sequential)
}
```

5 The latent class model

A latent class model supposes that each respondent belongs to one of K unobserved classes, and that **within a class the items are statistically independent** (the local-independence assumption). Writing π_k for the share of the population in class k , and ρ_{jck} for the probability that a member of class k gives response c to item j , the probability of observing respondent i 's full pattern of answers is a mixture over the classes:

$$P(\mathbf{y}_i) = \sum_{k=1}^K \pi_k \prod_{j=1}^J \rho_{j y_{ij} k}, \quad \text{with } \sum_{c=1}^{C_j} \rho_{jck} = 1 \text{ for every item } j \text{ and class } k.$$

The π_k are the **inclusion (class-size) probabilities** and the ρ_{jck} are the **conditional (item-response) probabilities**; together they are the *measurement model*. Each item is treated as an unordered (nominal) categorical variable, which is exactly what **poLCA** assumes, so the benchmark in Section 12 compares like with like. The items may have different numbers of categories; the model handles that without any special configuration. Because the items are independent within a class, a respondent who is missing an item still contributes through the items they did answer, which is what lets the model retain everyone for scoring without imputation.

Why this section matters: it is the contract the rest of the script honors. The local-independence assumption is what makes the classes interpretable as a measurement model rather than an arbitrary clustering, and it is the same assumption the indicator screen (Section 11) and the partial-response scoring (the scoring section) rely on; if it fails badly, the diagnostics there are where it shows up.

5.1 Why ordinary poLCA is not enough

poLCA maximizes the *unweighted* likelihood, so it estimates the class structure of the **achieved sample**. Under a complex design with unequal selection probabilities, the sample over- or under-represents parts of the population, and the unweighted estimates inherit that distortion. To describe the **population**, each respondent's contribution to the likelihood must be weighted by their base weight w_i , giving the *design-weighted pseudo-log-likelihood*

$$\ell_w(\theta) = \sum_i w_i \log \left(\sum_{k=1}^K \pi_k \prod_j \rho_{j y_{ij} k} \right).$$

Maximizing ℓ_w yields *pseudo-maximum-likelihood* estimates of the finite-population parameters. This is the standard design-based device for latent class models under complex sampling (Patterson, Dayton, and Graubard 2002, *Journal of the American Statistical Association* 97(459):721–728), and it is what we implement below, because **poLCA** has no argument for weights.

6 The data: LAPOP Mexico 2023

Everything dataset-specific happens in `prepare_data_lapop.R`, sourced above the configuration. That adapter reads the labelled .sav with `sjlabelled`, recodes nonresponse to `NA` by matching value labels (No sabe, No responde, No aplica) and prints an audit of exactly which codes were caught, authors English question wording and response labels onto the items (the dictionary keeps the extracted Spanish beside the English for confirmation), recodes the demographics, and applies the missingness policy: listwise deletion on design variables, demographics, and outcomes; item-partial respondents kept for the scoring section; only rows missing every item dropped. Adapting to another survey means writing a sibling adapter and pointing the config at its outputs; nothing else changes.

```
# The item dictionary: .sav code, Spanish question as extracted, authored
# English. CONFIRM the pairing here before trusting any downstream labels.
knitr::kable(lapop_dictionary,
  col.names = c("Item", ".sav code", "Question (Spanish, extracted)",
    "Question (English, authored)"),
  caption = "Item dictionary. The English wording is set onto the data
  ↪ and is what the class labeler will read; the Spanish column is
  ↪ the extracted source of truth to verify it against.")
```

Table 1: Item dictionary. The English wording is set onto the data and is what the class labeler will read; the Spanish column is the extracted source of truth to verify it against.

Item	.sav code	Question (Spanish, extracted)	Question (English, authored)
justice_system	b10a	Confianza en el sistema de justicia	To what extent do you trust the justice system?
electoral_tribunal	b11	Confianza en el Tribunal Supremo Electoral	To what extent do you trust the electoral tribunal?
armed_forces	b12	Confianza en las fuerzas armadas	To what extent do you trust the armed forces?
legislature	b13	Confianza en la legislatura	To what extent do you trust the legislature?
public_ministry	b15	Confianza en el Ministerio Público	To what extent do you trust the public ministry?
police	b18	Confianza en la policía nacional	To what extent do you trust the police?
auditor	b19	Confianza en la Auditoría Superior de la Federación	To what extent do you trust the auditor?
political_parties	b21	Confianza en los partidos políticos	To what extent do you trust the political parties?
president	b21a	Confianza en el presidente	To what extent do you trust the president?
supreme_court	b31	Confianza en el tribunal supremo	To what extent do you trust the supreme court?
municipality	b32	Confianza en su municipalidad	To what extent do you trust the municipality?
media	b37	Confianza en los medios de comunicación	To what extent do you trust the media?
elections	b47a	Confianza en las elecciones	To what extent do you trust the elections?

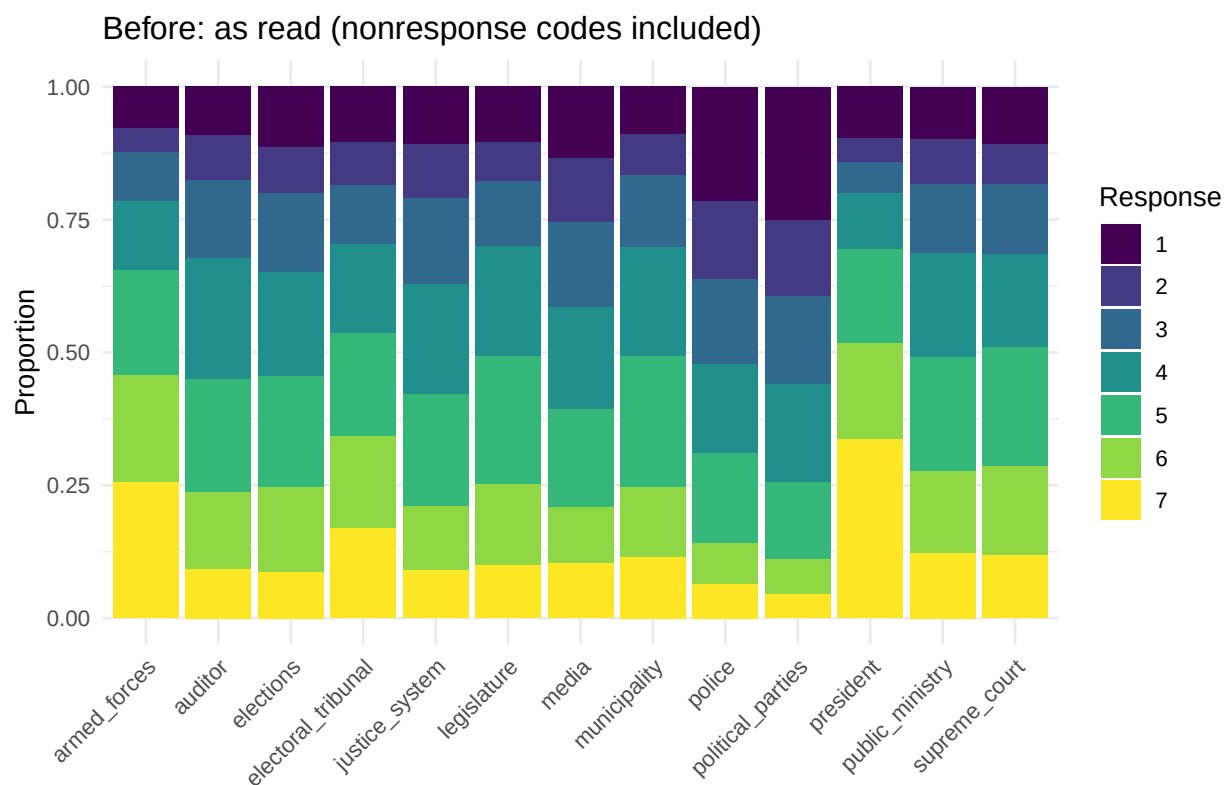
```
knitr::kable(lapop_responses, col.names = c("Value", "Response (English)"),
  caption = "Response scale for all thirteen items (LAPOP B-series
  ↪ anchors).")
```


Table 2: Response scale for all thirteen items (LAPOP B-series anchors).

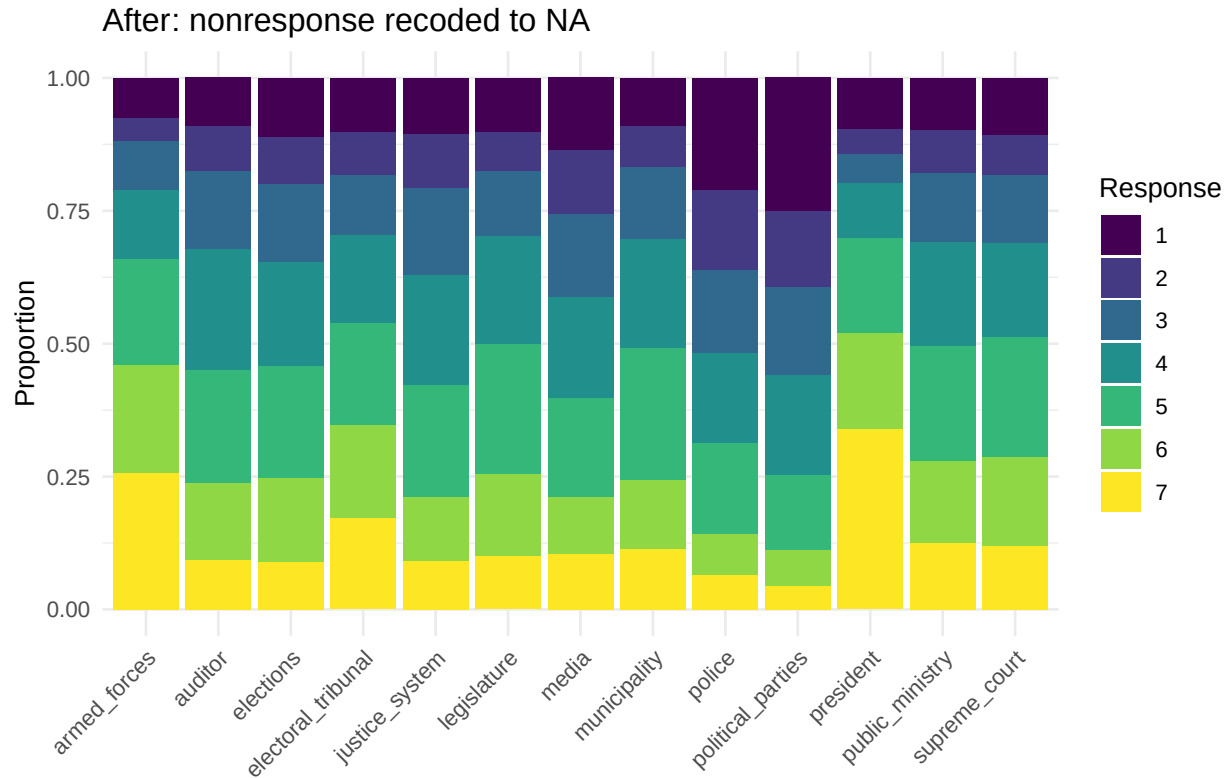
Value	Response (English)
1	Not at all
2	2
3	3
4	4
5	5
6	6
7	A lot

The two figures below show the response distributions before and after the nonresponse recode: the difference is exactly the mass in the printed NA audit, and the after panel is the 1..7 substantive scale the model sees.

```
# Printed sequentially on purpose: no patchwork dependency (its `/' operator
# breaks against S7-based ggplot2 versions).
print(plot_item_stack(lapop_before, lapop_items, "Before: as read (nonresponse
  ↪ codes included)"))
```



```
print(plot_item_stack(lapop, lapop_items, "After: nonresponse recoded to NA"))
```



```
# How the labeler will see each item: the exact structure item_meta() hands to
# the prompt builder (question wording + response labels). Confirm it item by
# item before any LLM call.
cats_preview <- map_int(cfg$items, ~ max(as.numeric(cfg$data[[.x]]), na.rm =
  TRUE)) |>
  set_names(cfg$items)
meta_preview <- item_meta(cfg$data, cfg$items, cats_preview, cfg$na_codes)
walk(cfg$items, function(it) {
  m <- meta_preview[[it]]
  cat(it, "\n Q: ", m$question,
    "\n Responses: ", paste(m$responses, collapse = " | "), "\n", sep = "")
})
```

```
justice_system
  Q: To what extent do you trust the justice system?
  Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
electoral_tribunal
  Q: To what extent do you trust the electoral tribunal?
  Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
armed_forces
  Q: To what extent do you trust the armed forces?
  Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
legislature
  Q: To what extent do you trust the legislature?
```

```

    Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
public_ministry
    Q: To what extent do you trust the public ministry?
    Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
police
    Q: To what extent do you trust the police?
    Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
auditor
    Q: To what extent do you trust the auditor?
    Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
political_parties
    Q: To what extent do you trust the political parties?
    Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
president
    Q: To what extent do you trust the president?
    Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
supreme_court
    Q: To what extent do you trust the supreme court?
    Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
municipality
    Q: To what extent do you trust the municipality?
    Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
media
    Q: To what extent do you trust the media?
    Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot
elections
    Q: To what extent do you trust the elections?
    Responses: Not at all | 2 | 3 | 4 | 5 | 6 | A lot

```

7 Preparing and validating the data

Before any modeling, the script confirms the indicators are well coded and the design columns are present, stopping with an explicit message if anything is wrong. This is the robustness layer: it converts silent downstream errors into clear, early ones. *Why this section matters: latent class estimates are only as trustworthy as the coding underneath them. A mislabeled category or an unflagged missing code does not error, it quietly biases every prevalence, so catching it here is what separates a reproducible result from a plausible-looking wrong one.*

“Don’t know” is treated as missing. Any code listed in `cfg$na_codes` is set to NA, and each item is then recoded to consecutive integers from 1 over whatever substantive categories it has, so items of different lengths are all handled the same way. The model is **fit on complete cases**, the respondents who answered every item, which keeps estimation clean and makes the **poLCA** benchmark in Section 12 an exact comparison. Respondents held out here because of a missing answer are not discarded: the scoring section assigns each of them a class from the fitted model using only the items they did answer. With a small fraction missing, this loses little and imputes nothing.

```

# Pull the working data and confirm required columns exist.
dat <- cfg$data
stopifnot(is.data.frame(dat))
required_cols <- c(cfg$items, cfg$aux, cfg$strata, cfg$psu, cfg$weight)
missing_cols <- setdiff(required_cols, names(dat))
if (length(missing_cols) > 0)
  stop("These configured columns are not in the data: ", paste(missing_cols,
    ↪ collapse = ", "))

# Design columns from .sav files arrive as haven_labelled, and vctrs forbids
# arithmetic on labelled vectors (the weighted EM computes w * posterior), so
# they are coerced to base types here, once, for any dataset: the weight to
# plain numeric unconditionally, strata and PSU stripped of the labelled class
# only if they carry it (they may legitimately be character elsewhere).
strip_labelled <- function(x) if (inherits(x, "haven_labelled")) as.numeric(x)
  ↪ else x
dat <- dat |>
  mutate(across(all_of(c(cfg$strata, cfg$psu)), strip_labelled),
    !!cfg$weight := as.numeric(.data[[cfg$weight]]))

# Map any configured missing/DK codes to NA, then recode each item to consecutive
# integers from 1 over its substantive responses (NA preserved). Coerce profiling
# covariates to factors so their levels drive the demographic domain estimates.
# (to_na() and recode_consecutive() are defined in survey_lca_source.R.)
dat_all <- dat |>
  mutate(across(all_of(cfg$items), ~ recode_consecutive(to_na(as.integer(.x)),
    ↪ cfg$na_codes))),
    across(all_of(cfg$aux), as.factor))

cats <- map_int(cfg$items, ~ max(dat_all[[.x]], na.rm = TRUE))
names(cats) <- cfg$items

# Complete-case subset for fitting, confidence intervals, and profiling.
  ↪ Respondents
# with any missing item are scored later (the scoring section) from their
  ↪ answered items.
complete_items <- stats::complete.cases(dat_all[, cfg$items, drop = FALSE])
dat_prepared <- dat_all[complete_items, , drop = FALSE]

# Report the per-item category counts and missing footprint.
tibble(item = cfg$items, categories = cats,
  n_missing = map_int(cfg$items, ~ sum(is.na(dat_all[[.x]])))) |>
  knitr::kable(align = "lrr")

```

item	categories	n_missing
justice_system	7	13

item	categories	n_missing
electoral_tribunal	7	8
armed_forces	7	10
legislature	7	41
public_ministry	7	33
police	7	2
auditor	7	108
political_parties	7	6
president	7	2
supreme_court	7	22
municipality	7	3
media	7	8
elections	7	9

Of 1580 respondents, 1430 answered every item and enter the fit; the remaining 150 are assigned a class in the scoring section.

8 The weighted EM estimator

The script estimates the measurement model with a weighted Expectation-Maximization (EM) algorithm. EM alternates two steps until the weighted log-likelihood stops improving.

The **E-step** computes each respondent's posterior probability of belonging to each class, given the current parameters:

$$p_{ik} = \frac{\pi_k \prod_j \rho_{j y_{ij} k}}{\sum_m \pi_m \prod_j \rho_{j y_{ij} m}}.$$

The **M-step** updates the parameters as *weighted* proportions, which is the only place the base weights enter:

$$\hat{\pi}_k = \frac{\sum_i w_i p_{ik}}{\sum_i w_i}, \quad \hat{\rho}_{jck} = \frac{\sum_i w_i p_{ik} \mathbf{1}(y_{ij} = c)}{\sum_i w_i p_{ik}}.$$

The $\hat{\rho}_{jck}$ denominator runs over respondents who answered item j ; because the model is fit on complete cases (Section 6), that is every respondent, so the sum is the same $\sum_i w_i p_{ik}$ written above. The implementation already restricts the sum correctly, which is what lets the same code score partial response patterns later.

Why this section matters: this is the engine. The weighting in the M-step is the single mechanism that moves every estimate from describing the achieved sample to describing the population, so its correctness propagates to every prevalence, profile, and interval downstream. Because the objective being maximized is the weighted *pseudo*-log-likelihood rather than an ordinary likelihood, these are *pseudo*-maximum-likelihood estimates. They are consistent for the population parameters, but, as

Section 10 explains, their standard errors cannot be read off the curvature of the objective in the usual way and must instead come from replication.

Two implementation choices make this robust and loop-free. First, the E-step is computed on the log scale and normalized with the **log-sum-exp** identity, which prevents numerical underflow when many small probabilities are multiplied together. Second, the iteration is expressed as a `purrr::reduce fold` over iteration indices rather than a `while` loop: each step takes the current parameter state and returns the updated one, and the fold stops once the weighted log-likelihood stops improving. The same E-step is reused, with the fitted parameters held fixed, to assign classes to respondents outside the fitting sample (the scoring section); because items are independent within a class, a missing item simply drops out of the per-respondent product, so a respondent is scored on whatever items they answered.

The estimator is implemented in `survey_lca_source.R`, where the code is annotated line by line: `em_run()` is the weighted EM fold, `make_inputs()` builds the integer item vectors and one-hot category matrices it consumes, `posterior_of()` is the standalone E-step reused for scoring, `align_to()` fixes label order across fits with the Hungarian algorithm, and `fit_lca()` runs many random starts and keeps the best weighted log-likelihood. Keeping the machinery there leaves this document focused on what the estimator means rather than on its bookkeeping.

9 Choosing the number of classes

The number of classes K is not known in advance and is chosen by comparing a sequence of models. Three quantities guide that choice, and because they are routinely misread, each is explained here in the terms you would use with an analyst.

BIC (Bayesian Information Criterion) scores a model by its fit, penalized for the number of parameters it uses:

$$\text{BIC} = -2\hat{\ell} + \text{df} \cdot \ln n, \quad \text{df} = (K - 1) + K \sum_j (C_j - 1).$$

The first term rewards fit (a higher log-likelihood $\hat{\ell}$ makes it smaller), and the second term charges a price for every additional parameter, a price that grows with the sample size n . **Lower BIC is better.** You read the column down the candidate values of K and look for the **minimum**; often the values fall steeply, reach a low point, and rise again, and that low point is the recommended number of classes. The interpretation is “the number of classes that explains the response patterns well without adding classes that only fit noise.” BIC is the primary tool for this decision in latent class analysis.

AIC (Akaike Information Criterion) has the same form with a smaller, fixed penalty (2 per parameter instead of $\ln n$):

$$\text{AIC} = -2\hat{\ell} + 2 \text{df}.$$

It is reported for completeness, but because its penalty is lighter it tends to **favor too many classes**, so when AIC and BIC disagree, prefer BIC for a parsimonious, interpretable solution.

Relative entropy answers a different question. BIC and AIC ask “how many classes?”; entropy asks “once we have them, how cleanly are respondents sorted into them?”

$$E = 1 - \frac{\sum_i \sum_k p_{ik} \ln p_{ik}}{n \ln K}.$$

It runs from 0 to 1. A value near 1 means most respondents have a posterior probability close to 1 for a single class, so the classification is sharp; a value near 0 means respondents are spread across classes with no clear home. A common rule of thumb treats values above about 0.8 as good separation. Entropy is **not** a measure of fit and must **not** be used to choose K on its own: a two-class model almost always has higher entropy than a five-class model simply because fewer classes are easier to separate. Read it alongside BIC as a description of how usable the resulting classification is, not as a selection criterion.

All enumeration here uses the **weighted** fit, so the chosen K describes the population, not the achieved sample. To keep the criteria honest, the weighted pseudo-log-likelihood is put on the sample-size scale before $\hat{\ell}$ enters BIC and AIC: the weights are normalized to sum to n , so the fit term and the $\ln n$ penalty are measured in the same units. Without this the fit term would be inflated by the average weight and the criteria would lean toward too many classes. The script reports all three quantities across the candidate range and selects the K that minimizes the weighted BIC. One standard companion to BIC is deliberately absent: the **bootstrapped likelihood ratio test** (BLRT), the usual complement for comparing K against $K + 1$ classes because the ordinary likelihood ratio statistic is not chi-square distributed for that comparison. The BLRT's parametric bootstrap simulates independent observations from the smaller model, and that premise does not carry over to a design-weighted pseudo-likelihood on a stratified, clustered sample, so applying it here would attach an invalid reference distribution to the decision. Enumeration therefore rests on the weighted BIC, read alongside entropy and the unweighted poLCA comparison in Section 12. *Why this section matters:* K is the one modeling choice with no default, and every later result is conditional on it, so a well-calibrated, defensible selection here is what keeps the prevalences and profiles from being artifacts of an over- or under-fit class structure.

```
# df_k() and entropy_R2() are defined in survey_lca_source.R.
w_vec <- dat_prepared[[cfg$weight]]

# Information criteria for a WEIGHTED pseudo-likelihood need the log-likelihood
#   and
# the sample-size penalty on the same scale. The base weights here do not sum to
# the sample size, so the weighted log-likelihood is rescaled to the n-scale
#   before
# forming AIC/BIC: multiplying by n / sum(w) is exactly fitting with weights
#   scaled
# to sum to n, because the pseudo-likelihood is linear in the weights. This
#   leaves
# the point estimates untouched (they are scale-invariant) and only calibrates
#   model
# selection, which otherwise leans toward too many classes.
scale_ic <- nrow(dat_prepared) / sum(w_vec)

enum <- furrr::future_map_dfr(cfg$K_range, function(K) {
  fit <- fit_lca(dat_prepared, w_vec, cats, cfg$items, K, cfg$n_starts)
  ll_n <- fit$ll * scale_ic # log-likelihood on the
# sum-to-n scale
  tibble(K = K,
    logLik = ll_n,
```

```

    df = df_k(K, cats),
    AIC = -2 * ll_n + 2 * df_k(K, cats),
    BIC = -2 * ll_n + df_k(K, cats) * log(nrow(dat_prepared)),
    entropy = entropy_R2(fit$post, K),
    converged = fit$converged)
}, .options = furrr::furrr_options(seed = TRUE,
  packages = c("matrixStats", "clue", "purrr", "dplyr", "tidyr")))

# K is chosen by minimum weighted BIC, unless cfg$K_force pins a specific number
# of classes. A forced K must be one of the fitted candidates in cfg$K_range so
# the enumeration table (and every downstream fit) actually contains it.
if (is.null(cfg$K_force)) {
  K_star <- enum |> slice_min(BIC, n = 1) |> pull(K) |> first()
} else {
  if (!(cfg$K_force %in% cfg$K_range))
    stop("cfg$K_force is ", cfg$K_force, ", which is not in cfg$K_range (",
      paste(range(cfg$K_range), collapse = ":"), "). Add it to cfg$K_range ",
      "or choose a value within it.")
  K_star <- as.integer(cfg$K_force)
}

enum |>
  mutate(across(c(logLik, AIC, BIC), ~ round(.x, 1)),
    entropy = round(entropy, 3)) |>
  knitr::kable(align = "r",
    caption = "Weighted fit statistics by number of classes. Selection
    ↪ is by minimum BIC.")

```

Table 4: Weighted fit statistics by number of classes. Selection is by minimum BIC.

K	logLik	df	AIC	BIC	entropy	converged
2	-31579.7	157	63473.3	64300.0	0.904	TRUE
3	-30176.2	236	60824.3	62067.0	0.913	TRUE
4	-29447.7	315	59525.5	61184.1	0.921	TRUE
5	-29047.7	394	58883.4	60957.9	0.907	TRUE
6	-28751.8	473	58449.5	60940.1	0.895	TRUE
7	-28537.7	552	58179.3	61085.8	0.893	TRUE
8	-28390.2	631	58042.5	61365.0	0.890	TRUE
9	-28242.7	710	57905.4	61643.8	0.890	TRUE
10	-28123.2	789	57824.4	61978.8	0.894	TRUE
11	-28024.0	868	57784.0	62354.4	0.902	TRUE
12	-27927.8	947	57749.6	62736.0	0.899	TRUE
13	-27856.1	1026	57764.2	63166.5	0.906	TRUE
14	-27758.2	1105	57726.4	63544.7	0.909	TRUE
15	-27687.9	1184	57743.9	63978.1	0.906	TRUE
16	-27598.9	1263	57723.9	64374.1	0.916	TRUE

The selected solution has **5 classes** (forced through `cfg$K_force`, overriding the BIC pick). The figure shows BIC and entropy across the candidate range; BIC is read for selection and entropy for separation.

```
enum |>
  dplyr::select(K, BIC, entropy) |>
  pivot_longer(-K, names_to = "criterion", values_to = "value") |>
  mutate(criterion = recode(criterion, BIC = "Weighted BIC", entropy = "Relative
    ↪ entropy")) |>
  ggplot(aes(K, value)) +
    geom_line(linewidth = 0.6, color = viridisLite::viridis(1, begin = 0.4)) +
    geom_point(size = 2.4, color = viridisLite::viridis(1, begin = 0.4)) +
    facet_wrap(~ criterion, scales = "free_y") +
    scale_x_continuous(breaks = cfg$K_range) +
    labs(x = "Number of latent classes (K)", y = NULL) +
    theme_lca()
```

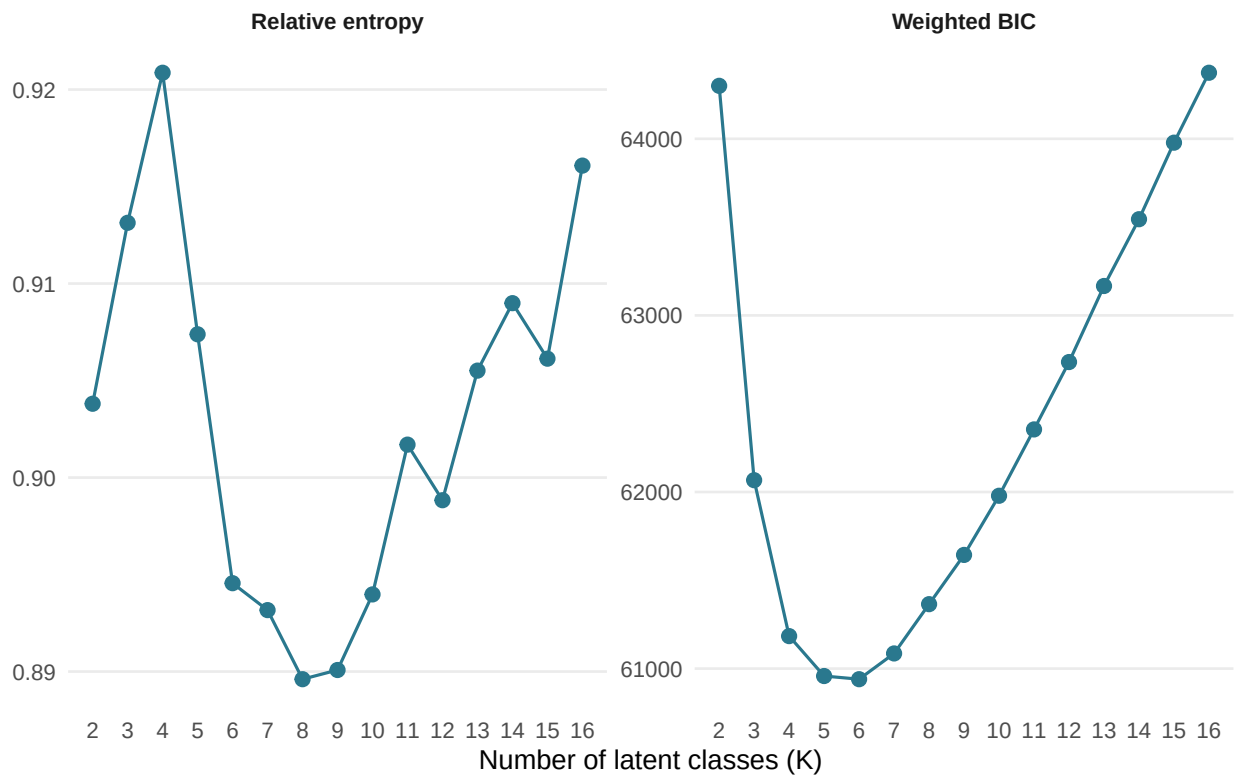


Figure 1: Weighted BIC and relative entropy across candidate numbers of classes. The chosen K minimizes BIC; entropy describes how cleanly respondents are classified.

10 The population measurement model

At the selected K , the script fits the model two ways on the same data: **weighted** (the population measurement model) and **unweighted** (which, fit with poLCA's assumptions, is what poLCA itself targets). The comparison in Section 12 then shows what the design weights change. *Why this section matters: this is the deliverable. The prevalences and the response-probability profiles are what a reader actually quotes and names, and fixing the label order here is what keeps every subsequent table, figure, and interval referring to the same classes.*

The model has two sets of parameters, and both deserve a plain reading because they are the substantive output of the analysis.

The **inclusion probabilities** π_k are the **class prevalences**: the fraction of the population in each class. They sum to 1 across classes. A prevalence of 0.35 for a class means an estimated 35 percent of the population belongs to that type. These are the headline numbers, the size of each segment.

The **conditional probabilities** ρ_{jck} are **item-response probabilities**: given that a respondent is in class k , the probability they give answer c to item j . For each class and item these sum to 1 across the answer categories. They are how you *characterize and name* a class. If, in one class, the probability of the most-open answer is high on every negotiation item, that class is the “open” type; if another class concentrates on the guarded answers, that is the “closed” type. Reading down the conditional probabilities for a class is reading its profile.

Class labels are arbitrary. The algorithm could number the classes in any order, so “Class 1” carries no meaning on its own. The script does not impose a substantive order on the classes: with items of differing lengths there is no single scale on which to rank them. What it does do is fix one label order (from the weighted fit) and align the unweighted and poLCA fits to it (Section 12), so that “Class 2” refers to the same class in every table and figure. You read each class by its conditional-probability profile; the next section drafts the names (LLM, or your CSV via `cfg$class_labels_csv`) and applies them to every downstream table and figure.

```
# Fit at K_star on the complete cases, weighted and unweighted. The weighted fit
↪ sets
# the label order; the unweighted fit is aligned to it so "Class k" is
↪ consistent.
fit_w <- fit_lca(dat_prepared, w_vec, cats, cfg$items, K_star, cfg$n_starts)
fit_u <- align_to(fit_lca(dat_prepared, rep(1, nrow(dat_prepared))), cats,
↪   cfg$items,
                                K_star, cfg$n_starts), fit_w)

# Posteriors recomputed from the FINAL parameters (the same kernel that scores
# out-of-sample respondents in the scoring section), then written back so the
↪ same replicate
# design serves both the measurement-model CIs and the profiling regression.
inp_full <- make_inputs(dat_prepared, cfg$items, cats)
post_w    <- posterior_of(fit_w$pi, fit_w$rho, inp_full$Y)
colnames(post_w) <- paste0("post_class", seq_len(K_star))
modal_int <- max.col(post_w, ties.method = "first")
dat_prepared <- dat_prepared |>
```

```
bind_cols(as_tibble(post_w)) |>
mutate(modal_class = factor(modal_int, levels = seq_len(K_star)),
       max_posterior = post_w[cbind(seq_len(n()), modal_int)])
```

11 Naming the classes

Class numbers carry no meaning, so this section attaches a short label and a one-to-two-sentence description to each class and applies them to **every downstream table and figure**. Three routes, in strict precedence. First, an **analyst CSV**: point `cfg$class_labels_csv` at a file with columns `K`, `Label`, `Description`; this requires `cfg$K_force`, because hand-written labels are only meaningful for a pinned number of classes, and when both are set the LLM never runs. Second, a **frozen LLM output**: once the LLM has run, its labels live in `class_labels_llm.csv` inside `cfg$out_dir` and are reloaded on every re-render, so the calls happen once; edit that file and point `cfg$class_labels_csv` at it to take over naming permanently. Third, the **LLM itself**: one call per class, drafting a label and a factual description from that class’s response probabilities and the item wording (extracted from `sjlabelled` attributes when the data are labelled, with value labels attached to `cfg$na_codes` dropped first so that 98/99-style “Don’t know” labels do not cost an item its substantive wording; item codes otherwise). An optional `cfg$survey_context`, one sentence naming country, year, and topic, is prepended so the model can resolve referents like “the agreement” in the item wording; when set, rule 1 gains a fence restricting the context to referent resolution, so nothing may be attributed to a class that its probabilities do not show, and when `NULL` the prompt is byte-identical to the certified context-free version.

The one-call-per-class design is empirical, not stylistic: in validation experiments, a single prompt carrying all classes systematically confused the closest class pairs, because a joint one-to-one assignment lets one confusion corrupt two classes, and a two-stage variant that fed a global comparison into each per-class call changed nothing. Isolated calls cost no accuracy and give one clean, independently frozen result per class. The provider is one OpenAI-compatible branch: `cfg$compass_base_url` with `cfg$llm_model`, your endpoint at work, OpenRouter at home, and no API key appears anywhere in this project (`.Rprofile/.Renv` supply it).

Two safeguards close the design. First, **harmonization**: isolation’s one blind spot is that two near-neighbor classes can draft the same label, since neither call saw the other. When a mechanical collision check fires (duplicate or heavily word-overlapping labels), one closing call sees only the finished label-description pairs, never the profiles, and edits **labels only**, minimally, anchored to each class’s own description; descriptions are frozen, non-colliding labels stay verbatim, and the pre-edit drafts are kept in a `Label_draft` column of the freeze file so every harmonizer change is auditable. When no collision exists, no extra call is made. Second, **determinism**: every class index is pinned by `cfg$seed` (each EM start re-seeds from it, so the classes come out in the same order on every run of the same data), and the same seed with temperature 0 is passed to the LLM. Cloud sampling is not bit-reproducible even so, which is exactly why the labels are frozen to CSV after the first run and reloaded thereafter: the freeze, not the seed, is what guarantees the class-label binding never drifts across renders.

Why this section matters: the labels are the words every reader will quote, and they are drafts, not findings. The LLM reads reliably but names loosely for near-neighbor or diffuse profiles, so verify

each label against its response profile in the next figure, and prefer editing the frozen CSV over trusting the first draft.

```
# get_class_labels() implements the precedence above (survey_lca_source.R,
# Section 4). The cache file freezes the LLM output for reproducibility.
class_labels <- get_class_labels(fit_w, dat_prepared, cfg$items, cats, cfg)

knitr::kable(class_labels,
              caption = "Class labels and descriptions in use: from
               ↪ cfg$class_labels_csv if supplied (requires K_force), otherwise
               ↪ the frozen LLM draft. Verify each against its response profile
               ↪ before quoting.")
```

Table 5: Class labels and descriptions in use: from `cfg$class_labels_csv` if supplied (requires `K_force`), otherwise the frozen LLM draft. Verify each against its response profile before quoting.

K	Label	Description
1	Moderate Institutional Trust	This class shows a strong tendency toward mid-to-high trust categories (4 and 5) across most institutions, with particularly high trust in the president, armed forces, and supreme court.
2	High Institutional Trust	This class is characterized by very high probabilities of selecting Category 7 across nearly all institutions, most notably for the president (0.82), public ministry (0.74), and armed forces (0.72).
3	Strong Institutional Trust	This class shows a strong lean toward higher trust categories (5, 6, and 7) across most institutions, with the highest probabilities concentrated in the president, armed forces, and supreme court.
4	Generalized Institutional Distrust	This class is characterized by a high probability of selecting Category 1 across all institutions, most notably for political parties (0.85) and police (0.76).
5	Selective Executive and Military Trust	This class is characterized by high distrust of political parties and police, contrasted with relatively high trust in the president and armed forces.

```
# Display labels for every downstream table and figure: short, wrapped, fixed
# class order so factor levels are stable everywhere.
# Three display tiers, because one label format cannot serve every surface:
#   lab_of      wrapped factor, for FIGURE legends and facets (newlines are fine
#   ↪ there)
#   lab_line    single line "Ck: Label", for SMALL tables (a newline inside a kable
#   ↪ cell breaks Typst layout)
#   lab_short  "Ck" only, for LARGE tables; the class-label table above is the
#   ↪ legend
lab_lev <- stringr::str_wrap(paste0("C", class_labels$K, ": ",
#   ↪ class_labels$Label), 22)
lab_of <- function(k) factor(lab_lev[k], levels = lab_lev)
```

```
lab_line <- function(k) paste0("C", k, ": ", class_labels$Label[k])
lab_short <- function(k) paste0("C", k)
```

12 Design-based confidence intervals via the stratified jackknife

Why this section matters: a prevalence without an interval invites false precision. The stratified jackknife is what turns the point estimates into defensible inference under the actual design, and it is the reason a typology built on few PSUs reports honestly wide intervals rather than misleadingly tight ones.

12.1 Why the survey weights alone are not enough

A natural question at this point is: the data already came with weights, so why not simply use them and be done? The answer is that survey weights do **one** of the two jobs that honest estimation requires, and the confidence intervals depend on the other.

The weights do the first job, **point estimation**: they make the prevalences and response probabilities describe the *population* rather than the achieved sample. That is precisely the weighted M-step in Section 7, $\hat{\pi}_k = \sum_i w_i p_{ik} / \sum_i w_i$, and the script does exactly that. If the only goal were the estimates themselves, “just use the weights” would be the complete answer.

The weights do **not** do the second job, **variance estimation**, because the precision of a survey estimate depends on *how the sample was selected*, not only on the weights. Two surveys can share an identical weight column yet differ greatly in precision: a sample of 1,200 people drawn as 20 clusters carries far less information than 1,200 people drawn independently, because respondents within a cluster tend to resemble one another, so each additional person in a cluster adds less new information than an independent person would. The weight attached to a respondent says nothing about which cluster they came from; the **design** (the stratum and PSU identifiers) is what carries that information. This is why the code below declares `svydesign(ids = ~psu, strata = ~stratum, weights = ~w)` and builds replicate weights from it, rather than working from the weight column alone.

The practical consequence of ignoring this is concrete and one-directional. If you weighted the data and then computed a standard error as though the weighted sample were a simple random sample of independent people, the estimate would be correct but the standard error would typically be **too small**, often by a factor of 1.5 to 2.5 in clustered samples. That inflation factor is the *design effect*. Using the weights for the estimate while ignoring the design for the variance, the single most common error in survey analysis, produces confidence intervals that are falsely narrow and overstates how much the data actually tell you.

There is a second, independent reason the variance cannot come from the weighted fit alone. The weighted objective $\sum_i w_i \log(\cdot)$ is a *pseudo*-log-likelihood, not a genuine likelihood for any probability model, because fractional and unequally weighted contributions do not correspond to a count of independent observations. The standard maximum-likelihood machinery that would ordinarily hand back standard errors from the curvature of the likelihood (the information matrix) therefore

does not apply. The variance has to be estimated by *replication*, refitting the model on systematically perturbed versions of the design, which is what the stratified jackknife does and what the next subsection describes.

In one sentence for a non-technical audience: **the weights tell us who the population is, and the design tells us how sure we are**, and we need both.

12.2 Estimating the design-based variance

Confidence intervals must reflect the sampling design, not pretend the data are a simple random sample. The script uses **replicate weights**: the design is declared, converted to a set of stratified jackknife (JKn) replicates, and the entire weighted EM is re-estimated on each replicate. The spread of the replicate estimates gives the design-based variance,

$$\widehat{\text{Var}}(\hat{\theta}) = \sum_r c_r (\hat{\theta}^{(r)} - \hat{\theta})^2,$$

where each JKn replicate r deletes one primary sampling unit and reweights the rest of its stratum, with the scale factor c_r that **survey** supplies. With about 20 PSUs across 9 strata this produces roughly 20 replicates and about 11 degrees of freedom for the variance, which is the true precision of the design regardless of the 1,200 respondents; intervals will be honestly wide.

Two robustness points are built in. The design declaration uses `nest = TRUE` so PSUs nested within strata are handled correctly. And **lonely PSUs** (a stratum with a single PSU, which would break JKn) are handled by setting `options(survey.lonely.psu = "adjust")`, which centers such a stratum's contribution at the grand mean rather than erroring; the script reports whether any were encountered. Each replicate refit is **warm-started at the full-sample solution and aligned to it**, which keeps the 20 refits stable, fast, and free of label switching.

```
options(survey.lonely.psu = "adjust")

# Report PSU structure so the analyst sees the degrees of freedom they actually
#   have.
# Note: dplyr::count() is namespace-qualified because matrixStats (loaded for
# rowLogSumExps) also exports a count(), and it loads after dplyr here.
psu_structure <- dat_prepared |>
  dplyr::distinct(.data[[cfg$strata]], .data[[cfg$psu]]) |>
  dplyr::count(.data[[cfg$strata]], name = "psus_in_stratum")
n_psu_total <- nrow(dplyr::distinct(dat_prepared, .data[[cfg$psu]]))
n_strata <- n_distinct(dat_prepared[[cfg$strata]])
lonely_any <- any(psu_structure$psus_in_stratum < 2)

# Declare the design and build stratified jackknife replicate weights.
des <- svydesign(ids = as.formula(paste0("~", cfg$psu)),
  strata = as.formula(paste0("~", cfg$strata)),
  weights = as.formula(paste0("~", cfg$weight)),
  data = dat_prepared, nest = TRUE)
rep_des <- as.svrepdesign(des, type = "JKn")
```

```

# Parameter bookkeeping as a TIBBLE, in the exact order flatten_fit() produces.
# This carries item/category/class as real columns, so nothing is recovered from
# string parsing later and the survey-question names are preserved downstream.
param_meta <- bind_rows(
  tibble(kind = "pi", item = NA_character_, category = NA_integer_, class =
    ↪ seq_len(K_star)),
  imap_dfr(cats, function(Cj, item_name)
    expand_grid(class = seq_len(K_star), category = seq_len(Cj)) |> # category
    ↪ varies fastest
    transmute(kind = "rho", item = item_name, category, class))
)
param_names <- param_meta |>
  mutate(nm = if_else(kind == "pi",
    sprintf("pi[%d]", class),
    sprintf("rho[%s,cat%d,class%d]", item, category, class)))
  ↪ |>
  pull(nm)
flatten_fit <- function(fit) set_names(c(fit$pi, unlist(map(fit$rho,
  ↪ as.vector)))), param_names)

# The statistic survey::withReplicates evaluates on each replicate: refit the
  ↪ weighted EM
# on the replicate weights, warm-started at the full-sample fit and aligned to
  ↪ it.
theta_fun <- function(w_rep, data) {
  inp <- make_inputs(data, cfg$items, cats)
  fit <- em_run(inp$Y, inp$OH, cats, w_rep, K_star,
    init = fit_w[c("pi", "rho")], maxit = 400L)
  flatten_fit(align_to(fit, fit_w))
}

rep_var <- withReplicates(rep_des, theta_fun)
est_vec <- flatten_fit(fit_w)
se_vec <- sqrt(diag(vcov(rep_var)))

# Tidy table of every probability with its design-based 95% CI:
  ↪ item/category/class
# are columns, ready to filter, join, and plot.
ci_tbl <- param_meta |>
  mutate(estimate = as.numeric(est_vec), se = as.numeric(se_vec),
    lo = pmax(0, estimate - 1.96 * se),
    hi = pmin(1, estimate + 1.96 * se))

```

The class-size (inclusion) probabilities, with their design-based intervals:


```

ci_tbl |>
  filter(kind == "pi") |>
  transmute(Class = lab_line(class),
            Prevalence = round(estimate, 3),
            SE = round(se, 4),
            `95% CI` = sprintf("%.3f, %.3f", lo, hi)) |>
  knitr::kable(align = "lrrr",
              caption = "Population class prevalences (inclusion probabilities)
              ↪ with design-based JKn confidence intervals. Labels are the
              ↪ drafted class names; verify each against its profile (next
              ↪ figure).")

```

Table 6: Population class prevalences (inclusion probabilities) with design-based JK_n confidence intervals. Labels are the drafted class names; verify each against its profile (next figure).

Class	Prevalence	SE	95% CI
C1: Moderate Institutional Trust	0.331	0.0201	[0.292, 0.371]
C2: High Institutional Trust	0.111	0.0103	[0.091, 0.132]
C3: Strong Institutional Trust	0.209	0.0191	[0.171, 0.246]
C4: Generalized Institutional Distrust	0.107	0.0114	[0.084, 0.129]
C5: Selective Executive and Military Trust	0.242	0.0205	[0.202, 0.282]

The item-response (conditional) probabilities are shown as a heatmap, which is far easier to scan than overlaid lines when there are several items: each panel is a class, each row an item, each column a response category, and the shade (and printed value) is the probability of that answer within that class. Reading a panel top to bottom gives that class's profile; comparing panels shows how the classes differ. Items with fewer categories simply leave the higher columns blank. The design-based confidence interval for every cell is in the exported table `lca_measurement_model_with_CIs.csv`; intervals are omitted from the heatmap only to keep it legible.

```

rho_plot <- ci_tbl |>
  filter(kind == "rho") |>
  mutate(Class = lab_of(class),
         item = factor(item, levels = cfg$items),
         category = factor(category),
         lbl = sprintf("%.2f", estimate),
         lbl_col = if_else(estimate > 0.55, "white", "grey15"))

ggplot(rho_plot, aes(category, fct_rev(item), fill = estimate)) +
  geom_tile(color = "white", linewidth = 0.5) +
  geom_text(aes(label = lbl, color = lbl_col), size = 2.4) +
  scale_color_identity() +
  facet_wrap(~ Class, nrow = 1) +
  scale_fill_viridis_c(limits = c(0, 1), name = "P(response | class)") +
  labs(x = "Response category", y = NULL) +

```



```
theme_lca() +
theme(panel.grid = element_blank(),
      legend.key.width = unit(1.4, "cm"))
```

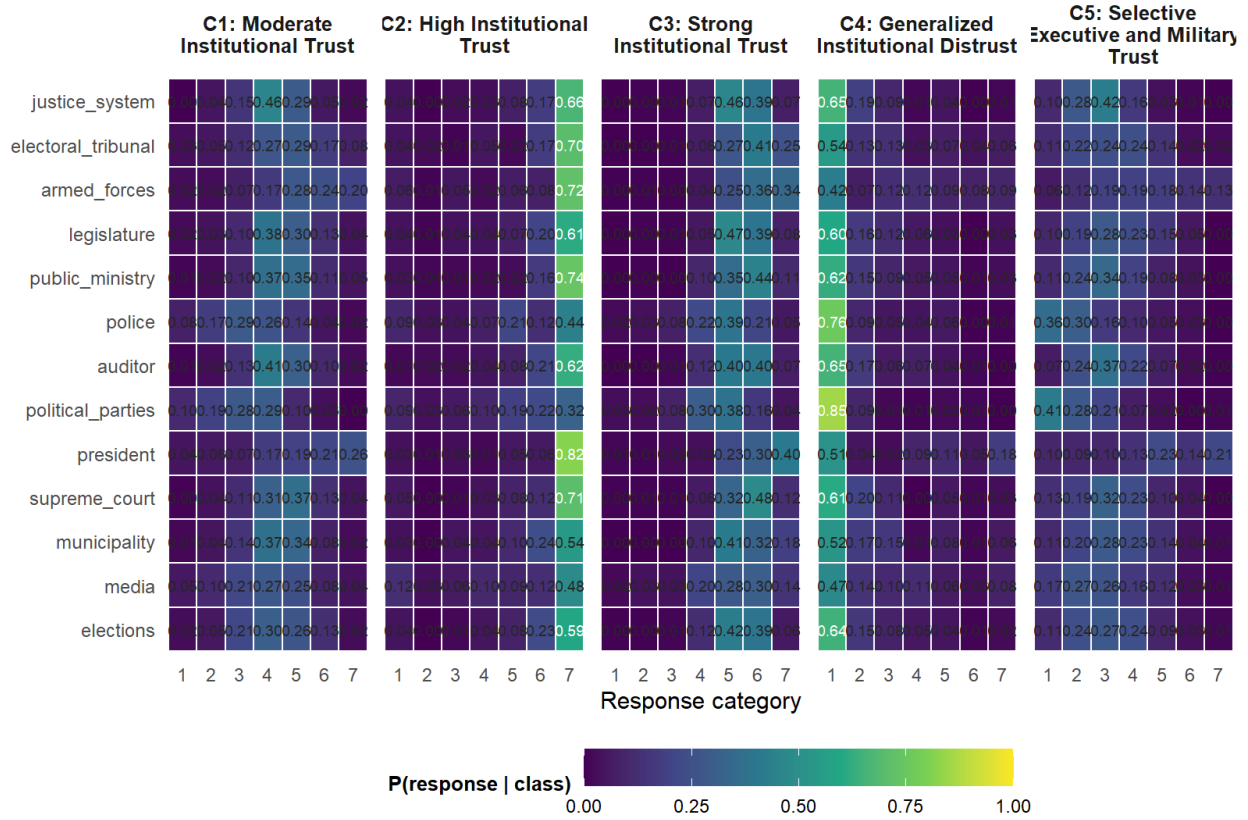


Figure 2: Population item-response (conditional) probabilities by class. Each panel is a class; rows are items, columns are response categories. Items of different lengths leave the higher columns blank. Cell shade and label give $P(\text{response} | \text{class})$. Panel headers carry the drafted class labels.

13 Which indicators carry the classification

Not every item earns its place in the model. An item helps define the classes only to the extent that the classes answer it *differently*; an item that every class answers the same way adds parameters and noise without sharpening the typology, and is a candidate to drop. The diagnostic below reads this directly off the fitted measurement model (the same ρ shown in the heatmap above).

For each item it computes a **discrimination** score: the average total-variation distance between the class-conditional response distributions, taken over all pairs of classes. For two classes a and b the total-variation distance on item j is $\frac{1}{2} \sum_c |\rho_{jca} - \rho_{jcb}|$, which is 0 when the two classes answer identically and 1 when their answers never overlap. Averaging over class pairs gives one number per item in $[0, 1]$: near 0 means the item does not separate the classes, larger means it does. The

low items are the ones to consider removing; after dropping them you refit and confirm the class structure and its interpretation are essentially unchanged.

The discrimination score is an effect size; on its own it does not say how small is “too small.” The automated screen below turns it into a keep-or-drop decision with BIC. For each item it asks whether letting that item’s response probabilities differ across classes is worth the extra parameters, by comparing the fitted model against a restricted model in which the item is made *class-independent* (its response distribution shared across classes, so it cannot inform the classes). When BIC prefers the restricted model, the item is flagged as a weak contributor. This is the model-comparison logic of Dean and Raftery (2010, *Annals of the Institute of Statistical Mathematics* 62(1):11–35) in a simplified form. Their method embeds the same full-versus-restricted BIC comparison inside a stepwise (headlong) search over which variables to keep; the screen here runs it once per item against the full model rather than as a search. The `LCAvarsel` package implements the further refinement of Fop, Smart, and Murphy (2017, *Annals of Applied Statistics* 11(4):2080–2110), which lets an excluded item depend on the other items instead of collapsing to its marginal as the simple restricted model here does. The size of the BIC gap grades the strength of the evidence: by Kass and Raftery (1995, *Journal of the American Statistical Association* 90(430):773–795), a difference beyond about 6 is strong and beyond about 10 very strong.

```
# Discrimination of each item under the fitted weighted model: how differently
  ↪ the classes
# answer it, measured as the mean over class pairs of the total-variation (TV)
  ↪ distance
# between their class-conditional response distributions. 0 = all classes answer
  ↪ alike (a
# drop candidate); near 1 = strong separation. Loop-free: map over items, then
  ↪ over pairs.
# Computed here; shown in the figure below and in the screen table further down.
class_pairs <- utils::combn(K_star, 2, simplify = FALSE) # list of c(a, b)
  ↪ class-index pairs

item_discrim <- tibble(
  item = cfg$items,
  discrimination = map_dbl(fit_w$rho, function(rho_j) {
    # rho_j is (categories x classes); rho_j[, k] is class k's response
    ↪ distribution.
    # TV distance for one pair of classes, then averaged over all pairs.
    mean(map_dbl(class_pairs, ~ 0.5 * sum(abs(rho_j[, .x[1]] - rho_j[, .x[2]])))
  })
) |>
  arrange(desc(discrimination))
```

```
item_discrim |>
  mutate(item = factor(item, levels = rev(item))) |>
  ggplot(aes(discrimination, item)) +
  geom_col(fill = viridisLite::viridis(1, begin = 0.45), width = 0.7) +
  geom_vline(xintercept = 0.1, linetype = "dashed", color = "grey55") +
  scale_x_continuous(limits = c(0, 1)) +
```

```
labs(x = "Discrimination (mean pairwise total-variation distance)", y = NULL) +
theme_lca()
```

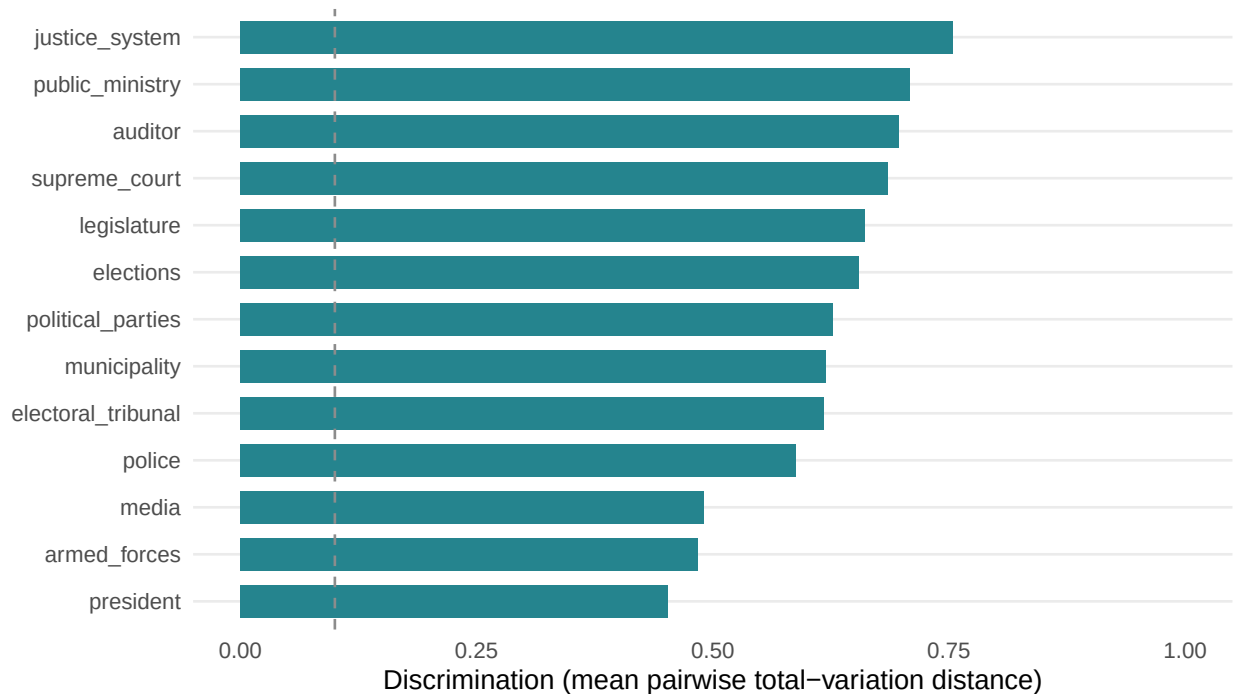


Figure 3: Item discrimination: how differently the latent classes answer each item (mean pairwise total-variation distance of the class-conditional response distributions). Bars near zero mark items that barely separate the classes and are candidates to drop before refitting. The dashed line at 0.1 is a soft reference, not a hard cutoff.

```
# Automated screen: does each item earn its class-DEPENDENT parameters by BIC?
  ↳ Compare, on
# the SAME respondents, the fitted model against a restricted model in which item
  ↳ j is made
# class-INDEPENDENT (its response distribution shared across classes, so it tells
  ↳ us nothing
# about the latent classes). Under local independence that restricted model
  ↳ factorises into
# an LCA on the OTHER items times item j's marginal, so its weighted
  ↳ pseudo-log-likelihood
# is ll(LCA without item j) + ll(item j's weighted marginal), and its BIC is
  ↳ directly
# comparable to the full model's. delta_bic = BIC(restricted) - BIC(full):
  ↳ NEGATIVE means
# BIC prefers dropping item j's class-dependence, so the item is flagged.
#
# This is the slowest chunk: it refits the LCA once per item. Lower cfg$n_starts
  ↳ here if you
```

```

# need speed, at some risk of a local optimum. It is reproducible because
  ↪ fit_lca() seeds
# each start from cfg$seed.
n_eff <- nrow(dat_prepared)
# Put every log-likelihood on the same sum-to-n scale used for the criteria in
# Section 8 (scale_ic = n / sum(w_vec)), so the screen's BIC comparison is
# calibrated the same way as the K-selection BIC.
ll_full <- fit_w$ll * scale_ic
df_full <- df_k(K_star, cats) # df_k() was defined in Section
  ↪ 8

# Weighted log-likelihood of one item under a single (class-independent)
  ↪ distribution.
marginal_ll <- function(item) {
  y <- as.integer(dat_prepared[[item]])
  p <- as.numeric(xtabs(w_vec ~ y)); p <- p / sum(p) # weighted response
  ↪ proportions, 1..C
  sum(w_vec * log(p[y]))
}

bic_screen <- tibble(item = cfg$items) |>
  mutate(delta_bic = furrr::future_map_dbl(cfg$items, function(j) {
    keep <- setdiff(cfg$items, j)
    fit_wo <- fit_lca(dat_prepared, w_vec, cats[keep], keep, K_star,
  ↪ cfg$n_starts)
    ll_restr <- (fit_wo$ll + marginal_ll(j)) * scale_ic # sum-to-n
    ↪ scale, as above
    df_restr <- df_k(K_star, cats[keep]) + (cats[[j]] - 1) # item j: one
    ↪ shared distribution
    (-2 * ll_restr + df_restr * log(n_eff)) - (-2 * ll_full + df_full *
    ↪ log(n_eff))
  }, .options = furrr::furrr_options(seed = TRUE,
    packages = c("matrixStats", "clue", "purrr", "dplyr", "tidyr"))))

# Combine the effect size and the decision into one ranked, flagged table.
item_screen <- item_discrim |>
  left_join(bic_screen, by = "item") |>
  mutate(flag = if_else(delta_bic < 0, "drop candidate", "keep")) |>
  arrange(delta_bic)

item_screen |>
  mutate(discrimination = round(discrimination, 3), delta_bic = round(delta_bic,
  ↪ 1)) |>
  knitr::kable(align = "lrrl",
    col.names = c("Item", "Discrimination", "Delta BIC", "Screen"),
    caption = "Automated indicator screen, most-droppable first.
  ↪ Discrimination is the mean pairwise total-variation distance
  ↪ (effect size). Delta BIC is BIC(item made class-independent)
  ↪ minus BIC(full model); a negative value means BIC prefers
  ↪ dropping the item's class-dependence, so it is flagged. The
  ↪ more negative the gap, the stronger the case (beyond about 6
  ↪ is strong, beyond 10 very strong).")

```

Table 7: Automated indicator screen, most-droppable first. Discrimination is the mean pairwise total-variation distance (effect size). Delta BIC is BIC(item made class-independent) minus BIC(full model); a negative value means BIC prefers dropping the item’s class-dependence, so it is flagged. The more negative the gap, the stronger the case (beyond about 6 is strong, beyond 10 very strong).

Item	Discrimination	Delta BIC	Screen
president	0.452	320.0	keep
armed_forces	0.484	424.0	keep
media	0.490	490.7	keep
police	0.588	742.1	keep
electoral_tribunal	0.618	827.5	keep
political_parties	0.627	863.6	keep
municipality	0.620	890.7	keep
legislature	0.661	987.7	keep
elections	0.654	988.1	keep
supreme_court	0.685	1116.4	keep
public_ministry	0.708	1161.9	keep
auditor	0.697	1197.1	keep
justice_system	0.754	1419.3	keep

Treat a flag as a prompt, not a verdict. An item can be statistically weak yet worth keeping for face validity, and a binary item tends to score lower simply because it has less room to separate classes. Drop the items that are both flagged and substantively dispensable, set `cfg$items` to the survivors, refit, and confirm the prevalences and class profiles are essentially unchanged; if they move a lot, the item was carrying real information and belongs in the model.

13.1 Local independence: the bivariate residual

The screen above asks whether each item *contributes* to the classes. A different failure is two items being *locally dependent*: more associated with each other than shared class membership explains, which violates the conditional-independence assumption the whole model is built on. The **bivariate residual (BVR)** diagnoses this per item pair. For items a and b it compares the weighted observed two-way table with the table the fitted model implies,

$$p_{rs}^{\text{exp}} = \sum_{k=1}^K \pi_k \rho_{a,rk} \rho_{b,sk},$$

as a Pearson X^2 scaled by n and divided by the table’s $(C_a - 1)(C_b - 1)$ degrees of freedom, so pairs with different category counts are comparable.

Two honesty notes before the table. First, under a weighted pseudo-likelihood on a complex design the usual chi-square reference for X^2 does not hold, so the BVR here is a **descriptive ranking index, not a test**; read the *relative* sizes. Absolute cutoffs from the simple-random-sampling

literature (values above roughly 3) do not transfer here, because unequal weights inflate the index above its nominal scale by roughly the weight-variation design effect; the operative signal is a pair standing well above the rest of its own table. Second, the residual is computed against the fitted K -class model, so it inherits that choice: a pair can look dependent simply because K is too small.

```
# bvr_pairs() is defined in survey_lca_source.R; fit$rho is positional in
  ↪ cfg$items
# order, so the items are passed in exactly that order.
bvr_tab <- bvr_pairs(dat_prepared, w_vec, cfg$items, fit_w)

bvr_tab |>
  mutate(bvr = round(bvr, 2)) |>
  slice_head(n = 10) |>
  knitr::kable(align = "llrr",
    col.names = c("Item A", "Item B", "Pair df", "BVR"),
    caption = "Largest bivariate residuals (BVR), a per-pair
  ↪ local-independence index: the design-weighted observed two-way
  ↪ table against the model-implied table, as Pearson X2 /
  ↪ (n-scaled) per degree of freedom. Descriptive ranking only; no
  ↪ chi-square test applies under the weighted pseudo-likelihood.
  ↪ Pairs standing well above the rest of the table mark
  ↪ associations the classes do not explain; absolute SRS cutoffs
  ↪ do not transfer under unequal weights.")
```

Table 8: Largest bivariate residuals (BVR), a per-pair local-independence index: the design-weighted observed two-way table against the model-implied table, as Pearson X^2 / (n-scaled) per degree of freedom. Descriptive ranking only; no chi-square test applies under the weighted pseudo-likelihood. Pairs standing well above the rest of the table mark associations the classes do not explain; absolute SRS cutoffs do not transfer under unequal weights.

Item A	Item B	Pair df	BVR
public_ministry	auditor	36	5.75
armed_forces	legislature	36	5.53
justice_system	supreme_court	36	3.21
police	political_parties	36	2.81
president	supreme_court	36	2.48
president	elections	36	2.40
electoral_tribunal	public_ministry	36	2.36
police	municipality	36	2.32
justice_system	elections	36	2.19
armed_forces	police	36	2.14

How to act on it, in order of preference. A pair that stands far above the rest and is *substantively* redundant (two near-identical wordings of one question) is a case for dropping one of the pair;

rerun the indicator screen after, since removing one member changes the other's contribution. If the pair is substantively distinct, the dependence may instead be a sign that K is too small, so check whether the pair's BVR falls at $K + 1$ before removing anything; a residual that melts away with one more class was class structure, not item redundancy. Only if it persists and both items must stay for content reasons does it become a documented limitation, a known local-independence violation whose main cost is overstated classification certainty for respondents whose pattern rides on that pair. A flat BVR table with no outlying pairs is affirmative evidence for the conditional-independence assumption and worth stating in the write-up.

14 Benchmark against poLCA, and a validation check

Two comparisons with poLCA close the loop. First, a **validation**: when all weights are set to one, the weighted EM and poLCA maximize the identical likelihood, so their estimates must agree up to label order and convergence tolerance. Confirming this is the strongest evidence that the custom EM is implemented correctly. Second, a **substantive benchmark**: poLCA fit normally (unweighted) estimates the achieved-sample structure, and its gap from the weighted estimates measures how much the design matters for this typology. The script also reports the K that poLCA's own BIC would select, since informative sampling can change the apparent number of classes. *Why this section matters: it does double duty. The unit-weight check is the script's correctness test against an independent, trusted implementation, and the weighted-versus-poLCA gap is the headline quantity on real data, the visible imprint of the design on the class shares, which is what you keep when no ground truth exists.*

```
# poLCA needs a data frame of items coded 1..C with no NA (it uses na.rm).
polca_df <- dat_prepared |> dplyr::select(all_of(cfg$items)) |> as.data.frame()
polca_f <- as.formula(paste0("cbind(", paste(cfg$items, collapse = ", "), ") ~
  ↪ 1"))

# --- substantive benchmark: poLCA at K_star, unweighted ---
set.seed(cfg$seed + 7L)
pl <- poLCA(polca_f, data = polca_df, nclass = K_star, nrep = 10,
  maxiter = 5000, na.rm = TRUE, verbose = FALSE)
pl_aligned <- align_to(list(pi = as.numeric(pl$P),
  rho = map(pl$probs, t),
  post = pl$posterior), fit_w)

# --- poLCA's own BIC-selected K across the candidate range ---
polca_bic <- furrr::future_map_dfr(cfg$K_range, function(K) {
  set.seed(cfg$seed + 10L + K)
  m <- poLCA(polca_f, data = polca_df, nclass = K, nrep = 5,
    maxiter = 5000, na.rm = TRUE, verbose = FALSE)
  tibble(K = K, poLCA_BIC = m$bic)
}, .options = furrr::furrr_options(seed = TRUE, packages = "poLCA"))
polca_K <- polca_bic |> slice_min(poLCA_BIC, n = 1) |> pull(K) |> first()

# --- validation: weighted EM with all weights = 1 should match poLCA ---
```

```

fit_unit <- align_to(fit_lca(dat_prepared, rep(1, nrow(dat_prepared)), cats,
  ↪   cfg$items,
                                K_star, cfg$n_starts), fit_w)
max_pi_gap <- max(abs(fit_unit$pi - pl_aligned$pi))
max_rho_gap <- max(map2_dbl(fit_unit$rho, pl_aligned$rho, ~ max(abs(.x - .y))))

tibble(`Latent class` = lab_line(seq_len(K_star)),
  `Population (weighted)` = round(fit_w$pi, 3),
  `Sample (poLCA)` = round(pl_aligned$pi, 3),
  `Difference (pp)` = round((fit_w$pi - pl_aligned$pi) * 100, 1)) |>
  knitr::kable(align = "lrrr",
    caption = "Class prevalence: weighted (population) versus
  ↪   unweighted poLCA (achieved sample). The difference is the
  ↪   imprint of the design.")

```

Table 9: Class prevalence: weighted (population) versus unweighted poLCA (achieved sample). The difference is the imprint of the design.

Latent class	Population (weighted)	Sample (poLCA)	Difference (pp)
C1: Moderate Institutional Trust	0.331	0.336	-0.5
C2: High Institutional Trust	0.111	0.111	0.0
C3: Strong Institutional Trust	0.209	0.207	0.1
C4: Generalized Institutional Distrust	0.107	0.107	0.0
C5: Selective Executive and Military Trust	0.242	0.239	0.4

The validation check returns a maximum class-size discrepancy of 0.0049 and a maximum response-probability discrepancy of 0.0085 between the unit-weight EM and poLCA; values near zero confirm the EM is correct. The weighted BIC selected 5 classes, and poLCA's unweighted BIC selected 6.

```

bind_rows(
  tibble(Class = lab_of(seq_len(K_star)), prevalence = fit_w$pi,
    Estimator = "Weighted (population)"),
  tibble(Class = lab_of(seq_len(K_star)), prevalence = pl_aligned$pi,
    Estimator = "Unweighted (poLCA)")
) |>
  ggplot(aes(Class, prevalence, fill = Estimator)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  scale_fill_manual(values = wu_pal) +
  scale_y_continuous(limits = c(0, NA)) +
  labs(x = NULL, y = "Class prevalence", fill = NULL) +
  theme_lca()

```

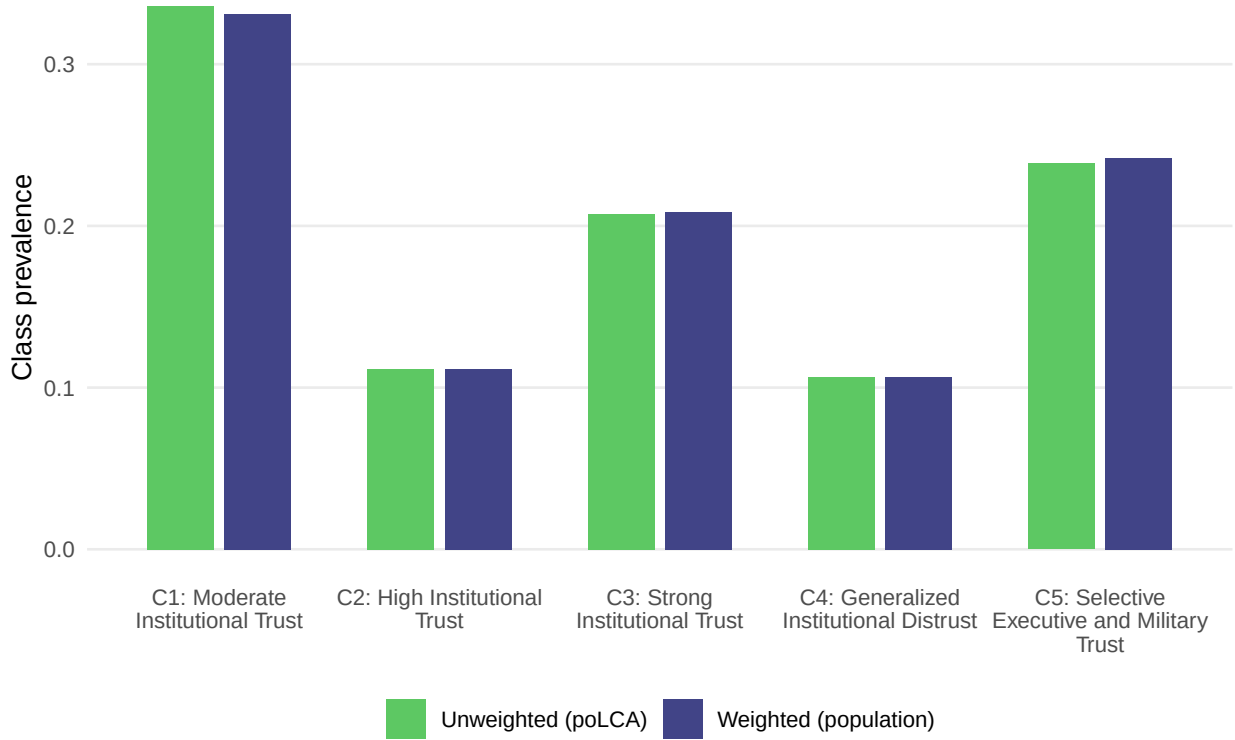



Figure 4: Population (weighted) versus achieved-sample (poLCA) class prevalences. Separation between the two indicates an informative design for this typology.

15 Assigning classes to respondents outside the fitting sample

The measurement model is a fixed set of parameters once it is estimated: the class sizes π_k and the item-response probabilities ρ_{jck} . Any respondent who answered the indicator items can be assigned a posterior probability of each class by applying the **same E-step** with those parameters held fixed,

$$p_{ik} = \frac{\pi_k \prod_j \rho_{j y_{ij} k}}{\sum_m \pi_m \prod_j \rho_{j y_{ij} m}},$$

and then taking the largest as their modal class. This is not a separate prediction model trained on demographics; it is the latent class model itself, used to *score* new response patterns, so it stays within the inferential framework. It applies to two groups: respondents held out of the fit here because they were missing an item, and entirely new respondents from a later wave or a separate file. Because the items are independent within a class, a missing answer drops out of the product, so a respondent is scored on whatever items they did answer; their maximum posterior will tend to be lower, reflecting that fewer items carry less information. *Why this section matters: it is what lets you fit on clean complete cases without throwing anyone away. Everyone receives a class from the same model, so the typology can be carried into later waves or merged files, and no one is dropped from the substantive story for a single missing answer.*

16.1 Which demographics belong here

Any demographic may appear here except the survey items that defined the classes. A variable used to build the survey weights, such as age, is fine to profile on and is not double-counting: the weights make each subpopulation estimate represent the population, and the demographic simply describes that represented population. The circularity to avoid is a different one: a variable used as an **indicator** of the latent classes must not also be profiled as if it described them, because that would explain the classes with the very items that built them. Demographics used in weighting are fine; the survey items that built the typology are not.

16.2 Design-based class prevalence by demographic

Rather than model class membership with a multinomial regression on the demographics, the profile reports, for each demographic level, the **design-based prevalence of each class within that subpopulation**: the share of that group estimated to belong to each type. This is a domain (subpopulation) estimate, the most standard quantity in survey sampling, and it is the right tool even when a demographic coincides with the design: a prevalence **within** a stratifying variable's levels is a well-defined quantity, whereas a regression coefficient on that variable would be confounded with the stratification and unstable.

The class-membership target is the **posterior probability** of each class, not a hard modal assignment. The design-weighted mean of the posterior within a subpopulation is that subpopulation's class prevalence, and because the design-weighted posterior mean over the whole sample reproduces the overall prevalences π_k from Section 9, the demographic breakdown is coherent with the headline numbers rather than attenuated by hard classification. Uncertainty is design-based from the same JK_n replicate weights used everywhere else: the prevalences are recomputed on each replicate and `survey::withReplicates` turns the spread into standard errors.

Why this section matters: this is the campaign-facing output. A class prevalence by demographic level, with a design-correct interval, is exactly what targeting needs (“this group is more this type than that group”), and computing it as a domain estimate keeps it stable and interpretable where a regression on a stratifying demographic would not be. The interval respects the design’s degrees of freedom, so a breakdown built on few PSUs per group reports honestly wide bands rather than false precision.

```
# The class-membership target is the soft posterior, already on dat_prepared and
# carried by the Section 10 replicate design (rep_des). post_cols are the K
# posterior columns in class order. The design objects des (the sampling design)
# and rep_des (its JKn replicates) are reused from Section 10 unchanged.
post_cols <- paste0("post_class", seq_len(K_star))

# Fixed layout for the flattened statistic: every (demographic, level, class)
  ↪ cell,
# levels in factor order, class varying fastest within a level. dom_meta names
  ↪ and
# orders the estimate vector so nothing is recovered by string parsing later.
dom_meta <- map_dfr(cfg$aux, function(v)
  expand_grid(plevel = levels(dat_prepared[[v]]), class = seq_len(K_star)) |>
```

```

    mutate(pvar = v)) |>
dplyr::select(pvar, plevel, class) |>
mutate(idx = row_number())

# The statistic survey::withReplicates evaluates on each replicate. For each
# demographic it forms a level-indicator matrix, then the weighted mean of each
# class posterior within each level: crossprod(M, w * P) / crossprod(M, w). The
# result is flattened to ONE named numeric vector in dom_meta's order
  ↪ (level-major,
# class-minor), because withReplicates needs a vector to build the covariance.
domain_theta <- function(w_rep, data) {
  P <- as.matrix(data[, post_cols, drop = FALSE])      # n x K, fixed across
  ↪ replicates
  wP <- w_rep * P                                     # n x K, weighted
  ↪ posteriors
  flat <- map(cfg$aux, function(v) {
    f <- factor(data[[v]], levels = levels(dat_prepared[[v]]))
    M <- model.matrix(~ f - 1)                        # n x L, one column per
  ↪ level
    num <- crossprod(M, wP)                           # L x K: weighted class
  ↪ sums per level
    den <- as.numeric(crossprod(M, w_rep))             # L: weighted total per
  ↪ level
    as.vector(t(num / den))                           # level-major,
    ↪ class-minor
  })
  unlist(flat, use.names = FALSE)
}

dom_rep <- withReplicates(rep_des, domain_theta)        # design-based covariance
  ↪ of every cell
dom_est <- domain_theta(weights(des), dat_prepared)    # full-sample point
  ↪ estimates
dom_V <- vcov(dom_rep)                                # full covariance, for
  ↪ the contrasts below
dom_se <- sqrt(diag(dom_V))
stopifnot(length(dom_est) == nrow(dom_meta))

# Per-level class prevalence with a design-based interval. The interval is formed
# on the logit scale and back-transformed so it stays inside [0, 1]; the standard
# error is mapped through the delta method, se_logit = se / (p (1 - p)).
  ↪ Estimates
# are clamped a hair off 0 and 1 only to keep the transform finite.
dom_tbl <- dom_meta |>
  mutate(p = as.numeric(dom_est), se = as.numeric(dom_se)) |>
  mutate(p_c = pmin(pmax(p, 1e-6), 1 - 1e-6),
         gse = se / (p_c * (1 - p_c)),

```

```

    lo = plogis(qlogis(p_c) - 1.96 * gse),
    hi = plogis(qlogis(p_c) + 1.96 * gse)) |>
dplyr::select(pvar, plevel, class, p, se, lo, hi)

```

```

# One compact table PER demographic: fits the page regardless of level-name
# length or K. The class-label table above is the legend for C1..CK.
walk(cfg$aux, function(v) {
  dom_tbl |>
  dplyr::filter(pvar == v) |>
  transmute(Level = plevel, Class = lab_short(class),
            cell = sprintf("%.2f [%.2f, %.2f]", p, lo, hi)) |>
  pivot_wider(names_from = Class, values_from = cell) |>
  knitr::kable(align = str_c("l", strrep("r", K_star)),
               caption = paste0("Class prevalence within ", v,
                                " (design-weighted mean posterior; 95% JKn
                                ↪ intervals on the logit scale). Each row
                                ↪ sums to one across the class columns.")) |>

  print()
  cat("\n\n")
})

```

Table 11: Class prevalence within prez_rating (design-weighted mean posterior; 95% JK_n intervals on the logit scale). Each row sums to one across the class columns.

Level	C1	C2	C3	C4	C5
18-29	0.36 [0.32, 0.40]	0.09 [0.07, 0.12]	0.21 [0.17, 0.24]	0.09 [0.07, 0.12]	0.25 [0.22, 0.28]
30-44	0.32 [0.28, 0.37]	0.10 [0.07, 0.13]	0.21 [0.17, 0.26]	0.11 [0.08, 0.14]	0.26 [0.21, 0.31]
45-59	0.32 [0.28, 0.37]	0.12 [0.09, 0.16]	0.17 [0.14, 0.22]	0.12 [0.09, 0.16]	0.26 [0.22, 0.30]
60+	0.30 [0.25, 0.36]	0.15 [0.11, 0.20]	0.26 [0.21, 0.31]	0.11 [0.08, 0.15]	0.18 [0.13, 0.24]

Table 12: Class prevalence within prez_rating (design-weighted mean posterior; 95% JK_n intervals on the logit scale). Each row sums to one across the class columns.

Level	C1	C2	C3	C4	C5
Female	0.34 [0.31, 0.38]	0.11 [0.09, 0.14]	0.19 [0.16, 0.22]	0.12 [0.10, 0.14]	0.24 [0.21, 0.27]
Male	0.32 [0.29, 0.35]	0.11 [0.09, 0.13]	0.23 [0.20, 0.26]	0.10 [0.08, 0.12]	0.24 [0.21, 0.28]

Table 13: Class prevalence within prez_rating (design-weighted mean posterior; 95% JK_n intervals on the logit scale). Each row sums to one across the class columns.

Level	C1	C2	C3	C4	C5
none	0.11 [0.03, 0.29]	0.41 [0.23, 0.62]	0.15 [0.05, 0.33]	0.21 [0.09, 0.42]	0.13 [0.04, 0.31]
primary	0.33 [0.28, 0.40]	0.17 [0.13, 0.22]	0.22 [0.16, 0.28]	0.09 [0.06, 0.14]	0.19 [0.14, 0.24]
secondary	0.34 [0.31, 0.37]	0.10 [0.08, 0.12]	0.22 [0.20, 0.25]	0.11 [0.09, 0.13]	0.23 [0.20, 0.26]
higher_ed	0.32 [0.27, 0.37]	0.07 [0.04, 0.11]	0.17 [0.13, 0.21]	0.10 [0.07, 0.14]	0.34 [0.29, 0.40]

Table 14: Class prevalence within prez_rating (design-weighted mean posterior; 95% JK_n intervals on the logit scale). Each row sums to one across the class columns.

Level	C1	C2	C3	C4	C5
rural	0.35 [0.30, 0.40]	0.12 [0.09, 0.16]	0.25 [0.20, 0.31]	0.09 [0.06, 0.14]	0.19 [0.15, 0.23]
urban	0.33 [0.30, 0.35]	0.11 [0.09, 0.13]	0.20 [0.17, 0.23]	0.11 [0.09, 0.13]	0.25 [0.23, 0.28]

Table 15: Class prevalence within prez_rating (design-weighted mean posterior; 95% JK_n intervals on the logit scale). Each row sums to one across the class columns.

Level	C1	C2	C3	C4	C5
Employed	0.33 [0.30, 0.36]	0.10 [0.08, 0.12]	0.20 [0.17, 0.23]	0.12 [0.10, 0.14]	0.24 [0.21, 0.27]
house_keeper	0.32 [0.27, 0.37]	0.13 [0.10, 0.17]	0.22 [0.18, 0.27]	0.09 [0.07, 0.12]	0.24 [0.20, 0.28]
Retired	0.34 [0.26, 0.43]	0.17 [0.10, 0.26]	0.19 [0.13, 0.27]	0.10 [0.05, 0.17]	0.20 [0.14, 0.29]
Student	0.38 [0.30, 0.47]	0.09 [0.05, 0.16]	0.25 [0.18, 0.34]	0.07 [0.03, 0.16]	0.21 [0.14, 0.29]
Unemployed	0.32 [0.24, 0.41]	0.09 [0.05, 0.15]	0.19 [0.12, 0.28]	0.09 [0.05, 0.16]	0.31 [0.23, 0.40]

Table 16: Class prevalence within prez_rating (design-weighted mean posterior; 95% JK_n intervals on the logit scale). Each row sums to one across the class columns.

Level	C1	C2	C3	C4	C5
Very satisfied	0.21 [0.15, 0.30]	0.33 [0.25, 0.43]	0.22 [0.15, 0.31]	0.05 [0.02, 0.11]	0.18 [0.12, 0.27]
Satisfied	0.35 [0.32, 0.38]	0.14 [0.11, 0.17]	0.26 [0.23, 0.30]	0.05 [0.04, 0.07]	0.19 [0.17, 0.22]

Table 16: Class prevalence within prez_rating (design-weighted mean posterior; 95% JK_n intervals on the logit scale). Each row sums to one across the class columns.

Level	C1	C2	C3	C4	C5
Dissatisfied	0.34 [0.31, 0.39]	0.04 [0.03, 0.06]	0.14 [0.11, 0.17]	0.15 [0.12, 0.18]	0.33 [0.29, 0.37]
Very dissatisfied	0.23 [0.16, 0.33]	0.05 [0.03, 0.11]	0.11 [0.06, 0.19]	0.34 [0.25, 0.43]	0.27 [0.19, 0.37]

Table 17: Class prevalence within prez_rating (design-weighted mean posterior; 95% JK_n intervals on the logit scale). Each row sums to one across the class columns.

Level	C1	C2	C3	C4	C5
Very good	0.30 [0.26, 0.35]	0.19 [0.16, 0.23]	0.23 [0.18, 0.28]	0.07 [0.05, 0.10]	0.21 [0.17, 0.25]
Good	0.34 [0.30, 0.37]	0.11 [0.08, 0.13]	0.25 [0.22, 0.29]	0.06 [0.05, 0.09]	0.24 [0.21, 0.27]
Neither good nor bad	0.38 [0.32, 0.44]	0.06 [0.04, 0.10]	0.13 [0.09, 0.18]	0.16 [0.13, 0.20]	0.27 [0.23, 0.31]
Bad	0.29 [0.20, 0.40]	0.00 [0.00, 0.01]	0.06 [0.03, 0.14]	0.30 [0.20, 0.41]	0.35 [0.25, 0.47]
Very bad	0.22 [0.13, 0.35]	0.06 [0.01, 0.25]	0.08 [0.03, 0.21]	0.47 [0.32, 0.62]	0.17 [0.09, 0.30]

```
# Horizontal orientation: long level names read cleanly on the y axis, and the
# dodge separates classes without angled-text tricks.
dom_tbl |>
  mutate(Class = lab_of(class)) |>
  ggplot(aes(p, plevel, color = Class)) +
  geom_pointrange(aes(xmin = lo, xmax = hi), size = 0.35, linewidth = 0.4,
                  position = position_dodge(width = 0.6)) +
  facet_wrap(~ pvar, scales = "free_y", ncol = 2) +
  scale_color_viridis_d(begin = 0.1, end = 0.8) +
  scale_x_continuous(limits = c(0, NA)) +
  labs(x = "Prevalence of class membership", y = NULL, color = NULL) +
  theme_lca()
```

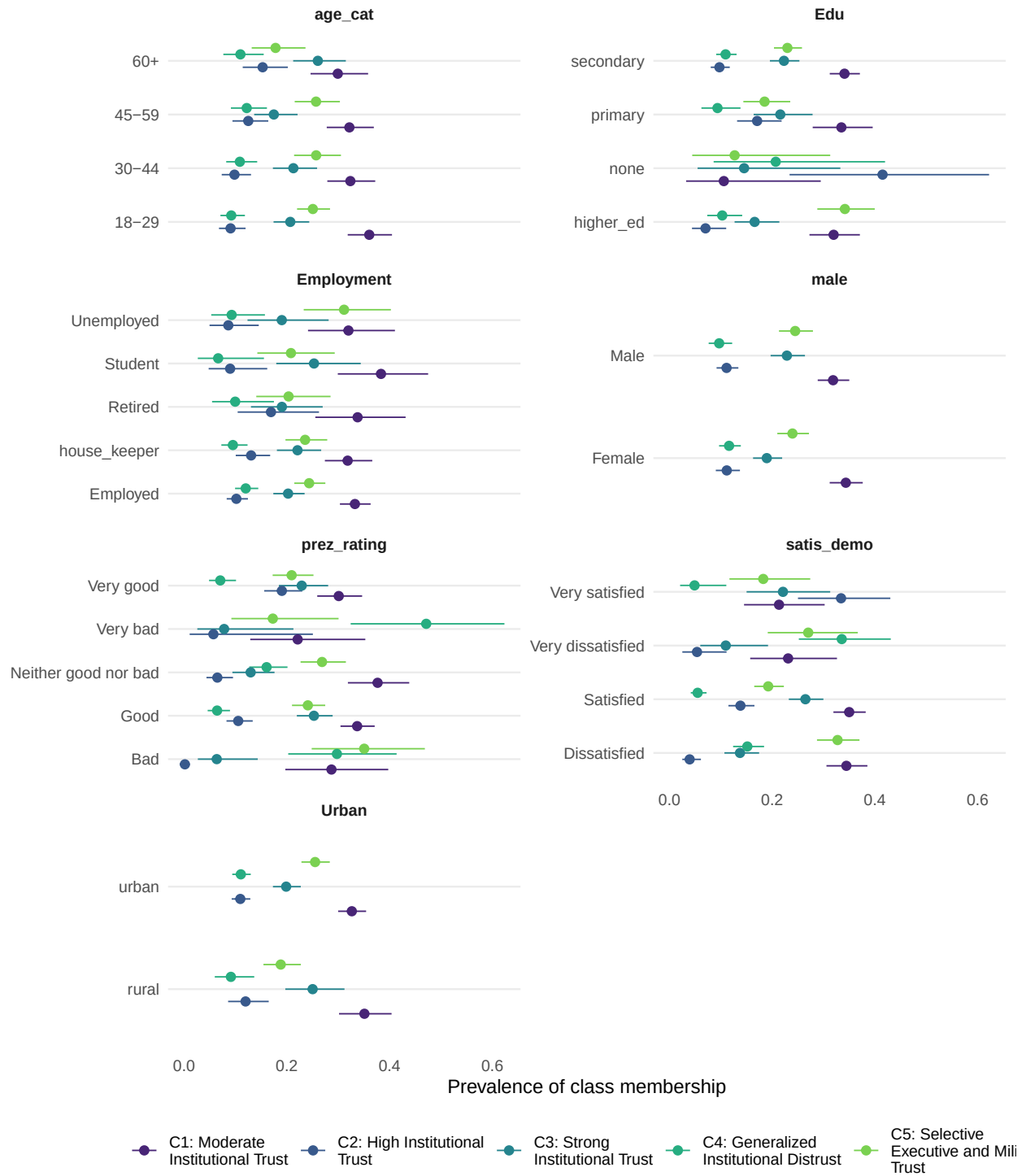


Figure 5: Design-based prevalence of each latent class within each demographic level, one panel per demographic, with 95% intervals from the JK_n replicate weights. Within a level the prevalences sum to one across classes. Wide bands reflect few PSUs in that group, not a coding artifact.

16.3 Differences between groups

For targeting, the operative quantity is often the **gap** between groups: how much more prevalent a class is at one demographic level than at a reference level. Each demographic's first level is the reference, and the difference in class prevalence (a risk difference on the probability scale) carries a design-based interval taken from the same replicate covariance, so the contrast sits on the identical variance footing as the prevalences above. A difference whose interval excludes zero is a group distinction the design can actually support; the interval width again reflects the available PSUs.

```
# Reference level = each demographic's first level. The difference in class
# prevalence between a level and its reference, with a design-based SE from the
# replicate covariance dom_V: Var(a - b) = Var(a) + Var(b) - 2 Cov(a, b). All
# vectorised through matrix indexing; no loops.
ref_idx <- dom_meta |>
  group_by(pvar, class) |>
  slice(1) |>
  ungroup() |>
  dplyr::select(pvar, class, ref = idx, ref_level = plevel)

dom_diff <- dom_meta |>
  inner_join(ref_idx, by = c("pvar", "class")) |>
  filter(plevel != ref_level) |>
  mutate(diff = dom_est[idx] - dom_est[ref],
         var = diag(dom_V)[idx] + diag(dom_V)[ref] - 2 * dom_V[cbind(idx, ref)],
         se = sqrt(pmax(var, 0)),
         lo = diff - 1.96 * se,
         hi = diff + 1.96 * se)

# These are DIFFERENCES, not probabilities: each cell is the previous table's
# prevalence at this level MINUS the same class's prevalence at the variable's
# reference (first) level, so negative values mean lower than the reference.
# One table per demographic, its reference level named once in the caption.
walk(cfg$aux, function(v) {
  d <- dom_diff |> dplyr::filter(pvar == v)
  if (nrow(d) == 0) return(invisible(NULL)) # single-level variables have no
    ↪ contrast
  ref_name <- d$ref_level[1]
  d |>
    transmute(Level = plevel, Class = lab_short(class),
              cell = sprintf("%.2f [% .2f, %.2f]", diff, lo, hi)) |>
    pivot_wider(names_from = Class, values_from = cell) |>
    knitr::kable(align = str_c("l", strrep("r", K_star)),
                 caption = paste0("Risk differences within ", v, ", each level
    ↪ minus the reference level \",
                                ref_name, "\" (NOT probabilities; negative =
    ↪ lower than the reference). An interval excluding zero is a distinction the
    ↪ design supports; JKn replicate intervals; classes C1..CK per the class-label
    ↪ table.)) |>
```

```

print()
cat("\n\n")
})

```

Table 18: Risk differences within prez_rating, each level minus the reference level “Very good” (NOT probabilities; negative = lower than the reference). An interval excluding zero is a distinction the design supports; JK_n replicate intervals; classes C1..CK per the class-label table.

Level	C1	C2	C3	C4	C5
30-44	-0.04 [-0.10, 0.03]	+0.01 [-0.03, 0.05]	+0.01 [-0.05, 0.06]	+0.02 [-0.02, 0.05]	+0.01 [-0.05, 0.06]
45-59	-0.04 [-0.10, 0.03]	+0.03 [-0.01, 0.08]	-0.03 [-0.08, 0.02]	+0.03 [-0.01, 0.07]	+0.01 [-0.04, 0.06]
60+	-0.06 [-0.13, 0.01]	+0.06 [0.01, 0.11]	+0.05 [0.00, 0.11]	+0.02 [-0.03, 0.06]	-0.07 [-0.13, -0.01]

Table 19: Risk differences within prez_rating, each level minus the reference level “Very good” (NOT probabilities; negative = lower than the reference). An interval excluding zero is a distinction the design supports; JK_n replicate intervals; classes C1..CK per the class-label table.

Level	C1	C2	C3	C4	C5
Male	-0.02 [-0.07, 0.02]	-0.00 [-0.03, 0.03]	+0.04 [-0.00, 0.08]	-0.02 [-0.05, 0.01]	+0.01 [-0.04, 0.05]

Table 20: Risk differences within prez_rating, each level minus the reference level “Very good” (NOT probabilities; negative = lower than the reference). An interval excluding zero is a distinction the design supports; JK_n replicate intervals; classes C1..CK per the class-label table.

Level	C1	C2	C3	C4	C5
primary	+0.23 [0.09, 0.37]	-0.24 [-0.44, -0.04]	+0.07 [-0.08, 0.22]	-0.11 [-0.29, 0.06]	+0.06 [-0.08, 0.19]
secondary	+0.23 [0.11, 0.36]	-0.32 [-0.52, -0.11]	+0.08 [-0.06, 0.21]	-0.10 [-0.27, 0.07]	+0.10 [-0.03, 0.23]
higher_ed	+0.21 [0.08, 0.35]	-0.34 [-0.55, -0.14]	+0.02 [-0.12, 0.16]	-0.10 [-0.27, 0.06]	+0.21 [0.07, 0.36]

Table 21: Risk differences within prez_rating, each level minus the reference level “Very good” (NOT probabilities; negative = lower than the reference). An interval excluding zero is a distinction the design supports; JK_n replicate intervals; classes C1..CK per the class-label table.

Level	C1	C2	C3	C4	C5
urban	-0.02 [-0.08, 0.03]	-0.01 [-0.05, 0.03]	-0.05 [-0.12, 0.01]	+0.02 [-0.02, 0.06]	+0.07 [0.02, 0.11]

Table 22: Risk differences within prez_rating, each level minus the reference level “Very good” (NOT probabilities; negative = lower than the reference). An interval excluding zero is a distinction the design supports; JK_n replicate intervals; classes C1..CK per the class-label table.

Level	C1	C2	C3	C4	C5
house_keeper	-0.01 [-0.07, 0.04]	+0.03 [-0.01, 0.07]	+0.02 [-0.03, 0.07]	-0.02 [-0.06, 0.01]	-0.01 [-0.06, 0.04]
Retired	+0.01 [-0.09, 0.10]	+0.07 [-0.01, 0.15]	-0.01 [-0.09, 0.06]	-0.02 [-0.08, 0.04]	-0.04 [-0.12, 0.04]
Student	+0.05 [-0.04, 0.14]	-0.01 [-0.07, 0.05]	+0.05 [-0.04, 0.14]	-0.05 [-0.11, 0.01]	-0.04 [-0.11, 0.04]
Unemployed	-0.01 [-0.10, 0.08]	-0.02 [-0.06, 0.03]	-0.01 [-0.09, 0.07]	-0.03 [-0.08, 0.03]	+0.07 [-0.02, 0.16]

Table 23: Risk differences within prez_rating, each level minus the reference level “Very good” (NOT probabilities; negative = lower than the reference). An interval excluding zero is a distinction the design supports; JK_n replicate intervals; classes C1..CK per the class-label table.

Level	C1	C2	C3	C4	C5
Satisfied	+0.14 [0.05, 0.22]	-0.20 [-0.29, -0.10]	+0.04 [-0.04, 0.13]	+0.01 [-0.04, 0.05]	+0.01 [-0.08, 0.10]
Dissatisfied	+0.13 [0.04, 0.22]	-0.29 [-0.39, -0.20]	-0.08 [-0.18, 0.01]	+0.10 [0.05, 0.15]	+0.14 [0.06, 0.23]
Very dissatisfied	+0.02 [-0.10, 0.13]	-0.28 [-0.38, -0.18]	-0.11 [-0.22, -0.00]	+0.29 [0.18, 0.39]	+0.09 [-0.03, 0.21]

Table 24: Risk differences within prez_rating, each level minus the reference level “Very good” (NOT probabilities; negative = lower than the reference). An interval excluding zero is a distinction the design supports; JK_n replicate intervals; classes C1..CK per the class-label table.

Level	C1	C2	C3	C4	C5
Good	+0.04 [-0.02, 0.09]	-0.08 [-0.13, -0.04]	+0.02 [-0.03, 0.08]	-0.01 [-0.04, 0.03]	+0.03 [-0.01, 0.08]
Neither good nor bad	+0.08 [0.00, 0.15]	-0.13 [-0.17, -0.08]	-0.10 [-0.16, -0.04]	+0.09 [0.04, 0.14]	+0.06 [-0.00, 0.12]
Bad	-0.01 [-0.12, 0.09]	-0.19 [-0.23, -0.15]	-0.17 [-0.24, -0.09]	+0.23 [0.12, 0.33]	+0.14 [0.02, 0.26]
Very bad	-0.08 [-0.20, 0.04]	-0.13 [-0.23, -0.03]	-0.15 [-0.25, -0.05]	+0.40 [0.24, 0.56]	-0.04 [-0.15, 0.08]

One limitation is stated plainly, as elsewhere in the script: these estimates treat the measurement model as given. The replicate weights carry the design variance of the prevalences and their differences, but they do not propagate the uncertainty in the latent class fit itself (the π_k and ρ_{jek} estimated in Section 9). Propagating that as well would mean refitting the whole model inside a design bootstrap, a heavier exercise not done here.

17 Exporting results

Why this section matters: it makes the analysis usable beyond the render. The enumeration, the design-based interval table, and the fully scored respondent file are the artifacts that feed manuscripts, slides, and downstream merges, and writing them out is what turns a one-time report into a reproducible dataset others can build on.

```
readr::write_csv(enum, file.path(cfg$out_dir, "lca_enumeration.csv"))
readr::write_csv(ci_tbl |>
  mutate(across(where(is.numeric), ~ round(.x, 5)),
    class_label = class_labels$Label[class]),
  file.path(cfg$out_dir, "lca_measurement_model_with_CIs.csv"))

readr::write_csv(class_labels, file.path(cfg$out_dir, "lca_class_labels.csv"))
readr::write_csv(dat_scored_all |>
  dplyr::select(any_of("id"), all_of(c(cfg$strata, cfg$psu,
    ↪ cfg$weight))),
  starts_with("post_class"), modal_class,
  ↪ max_posterior),
  file.path(cfg$out_dir, "lca_caselevel_assignments.csv"))
```

Four files are written to `cfg$out_dir`: the enumeration table, the full measurement model with design-based confidence intervals (one row per probability, with `item`, `category`, `class`, and the applied `class_label` as columns), the class-label sheet (K, Label, Description; edit it and point

`cfg$class_labels_csv` at it to take control of naming), and the case-level assignments for **all** respondents, including those scored outside the fit in the scoring section (posteriors, modal class, and maximum posterior), ready to merge back into the survey data.

18 Adapting this script to your survey: a checklist

1. **Write a dataset adapter.** Copy `prepare_data_lapop.R`, edit only its `pd` block (path, design-variable names, item codes and English names, nonresponse regex, wording template, demographic recodes), source it in the adapter chunk, and point `cfg$data/cfg$items` at its outputs. Nothing else in this document changes.
2. **Set `cfg$items`** to your indicator columns, each coded as consecutive integers from 1 over its substantive answers. Items may have **different numbers of categories**. Put any “Don’t know” or missing code in `cfg$na_codes` (or use `NA` directly); those responses are treated as missing, the model is fit on complete cases, and missing-item respondents are assigned a class in the scoring section. The category-count and missing table after Section 6 lets you confirm the coding.
3. **Set `cfg$strata`, `cfg$psu`, `cfg$weight`** to your design columns. This is the analogue of `svyset psu [pweight = w], strata(stratum)`. If your file ships **replicate weights** instead of strata and PSU identifiers, replace each `svydesign(...)` plus `as.svrepdesign(...)` pair with a single `svrepdesign(repweights = ..., weights = ..., type = ...)`; the rest of the variance code is unchanged.
4. **Set `cfg$aux`** to the demographics you want in the profile. Demographics used to build the weights (age, for instance) are fine and often helpful here; the only variables to exclude are the survey items that defined the classes.
5. **Optionally set `cfg$K_force`** to pin the number of classes when the weighted BIC prefers more than are interpretable. Leave it `NULL` to select K by BIC (Section 8); set it to an integer within `cfg$K_range` to override the pick. The enumeration table still prints either way, so the BIC evidence remains visible alongside the forced choice.
6. **Run, then read enumeration first.** Choose K on the weighted BIC, sanity-checked against entropy and interpretability (or pin it with `cfg$K_force`); the validation check should show the EM matching `poLCA` at unit weights before you trust the weighted output.
7. **Mind the degrees of freedom.** Your design-based intervals carry roughly (number of PSUs minus number of strata) degrees of freedom; with few PSUs they will be wide, and that width is honest.
8. **Read the demographic profile as design-based prevalences (the profiling section).** It reports each class’s prevalence within each demographic level, and the differences between groups, with design-based intervals from the JK_n replicate weights. Region is included even when it is the stratum, because a within-region prevalence is a domain estimate rather than a regression coefficient. Wide bands reflect few PSUs in a group, not an error. These estimates treat the measurement model as given and do not propagate the step-one latent class uncertainty.
9. **Keep `survey_lca_source.R` next to this document.** It holds the plotting and data-prep helpers, the weighted EM engine, and the LLM class-labeling functions, and is sourced in the setup chunk. The functions are dataset-agnostic, so the same file serves any study, but the render fails if it is not in the working directory.

10. **Score respondents outside the fit** with `score_lca(newdata, fit_w, cfg$items)` (the scoring section): it assigns a class to anyone carrying the item columns, including a fresh data frame, as long as those items are recoded the same way.
11. **Class labels.** By default the LLM drafts a label and description per class (one call per class, frozen to `class_labels_llm.csv` in `cfg$out_dir`), and every downstream figure and table uses them. To take control: edit that file, or write your own with columns `K`, `Label`, `Description`, point `cfg$class_labels_csv` at it, and set `cfg$K_force`; the LLM then never runs. At work set `cfg$compass_base_url` and `cfg$llm_model`, and set `cfg$llm_key_env` `<- NULL` so the `.Rprofile`-supplied key is used; the key itself never appears in any script.

```
future::plan(future::sequential) # close any parallel workers
```